

Channel Map-Based Statistical CSI Acquisition and Performance Analysis for HRLLC Systems

Yang Ou, *Student Member, IEEE*, Jie Huang[✉], *Senior Member, IEEE*, Shuaifei Chen[✉], *Member, IEEE*,
Cheng-Xiang Wang[✉], *Fellow, IEEE*, and El-Hadi M. Aggoune[✉], *Life Senior Member, IEEE*

Abstract—Hyper-reliable and low-latency communication (HRLLC) constitutes a pivotal application scenario for sixth-generation (6G) wireless communication systems, with its implementation contingent upon the quality of acquired channel state information (CSI). The channel map has emerged as a promising paradigm to enhance CSI accuracy by exploiting environmental geometry. In this paper, a channel map-based statistical CSI acquisition method is proposed, which leverages multipath information in environment to improve the CSI acquisition accuracy. Then, a general performance analysis framework applicable to HRLLC multiple-input multiple-output (MIMO) systems is established, jointly considering the pilot contamination, channel spatial correlation, channel aging, and channel map accuracy. Closed-form expressions for key performance metrics are derived, including normalized mean square error (NMSE) of minimum mean square error (MMSE) channel estimator and lower bounds on signal-to-interference-plus-noise ratio (SINR) with maximum ratio (MR) schemes. Based on that, significant effects of environmental parameters and channel statistical properties on HRLLC systems are revealed and further illustrated by simulations. Subsequently, the achievable data rate (ADR) of HRLLC systems utilizing different error-level channel maps are compared and analyzed, demonstrating performance enhancement from channel maps even when employing imperfect channel maps. The fundamental trade-off relationship between ADR and reliability is investigated, which is further improved through the block length optimization.

Index Terms—Statistical CSI acquisition, performance analysis, HRLLC, channel map, trade-off.

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Yang Ou, Jie Huang, and Cheng-Xiang Wang are with the National Mobile Communications Research Laboratory, School of Information Science and Engineering, Southeast University, Nanjing 211189, China, and also with Purple Mountain Laboratories, Nanjing 211111, China (e-mail: yang_ou@seu.edu.cn; j_huang@seu.edu.cn; chxwang@seu.edu.cn).

Shuaifei Chen is with Purple Mountain Laboratories, Nanjing 211111, China, and also with the National Mobile Communications Research Laboratory, School of Information Science and Engineering, Southeast University, Nanjing 211189, China (e-mail: shuaifeichen@seu.edu.cn).

El-Hadi M. Aggoune is with the AI and Sensing Technologies Research Center, University of Tabuk, Tabuk 47315, Saudi Arabia (e-mail: haggoune@ut.edu.sa).

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I. INTRODUCTION

THE sixth-generation (6G) wireless communication era is characterized by the vision of “global coverage, all spectra, full applications, all senses, all digital, and strong security” [1]. Hyper-reliable and low-latency communication (HRLLC) represents one of the cornerstone application scenarios of forthcoming 6G wireless systems [2]. Within the 6G paradigm, HRLLC encompasses a broad range of mission-critical applications, including autonomous driving, drone real-time control, industrial automation, extended reality, and intelligent transportation systems [3]. Compared to ultra-reliable and low-latency communication (URLLC) in fifth-generation (5G) systems, HRLLC in the 6G era demands superior data throughput, further reduced latency, and substantially enhanced reliability [4]. Among the various enabling technologies, finite-blocklength (FBL) transmission has emerged as the most promising approach for meeting these stringent performance requirements [5]. Tight bounds on the maximum achievable rate of FBL transmission were established as functions of block length, error probability, and signal-to-noise ratio (SNR) in [6], and subsequently extended to non-asymptotic upper and lower bounds over Rayleigh and Rician block-fading channels in [7]. Although FBL transmission theory commonly assumes an idealized Gaussian codebook to establish fundamental performance limits, it provides an indispensable theoretical benchmark for evaluating practical transmission schemes in HRLLC scenarios. To investigate gaps between theoretical capacity bounds and practical implementations, traditional CSI acquisition for turbo-coded orthogonal frequency-division multiplexing (OFDM) systems over fading channels were studied in [8], [9], and [10].

Massive multiple-input multiple-output (MIMO) technology has been established as a powerful methodology for enhancing spectral efficiency (SE), and has been identified as a key enabler of HRLLC [11]. In multi-user cellular networks, the accuracy of pilot-based channel estimation is degraded by inter-user interference, giving rise to the well-known pilot contamination problem [12]. A performance analysis framework for URLLC systems accounting for pilot contamination in multi-user massive MIMO systems over Rayleigh and Rician fading channels was presented in [13]. An efficient methodology for evaluating theoretical bounds on HRLLC error probability over memoryless Rayleigh block-fading channels in MIMO networks was proposed in [14]. The performance of multi-user massive MIMO systems over

spatially correlated Rician fading channels was investigated in [15]. The performance of time-space exchange technology under FBL transmission was analyzed in [16], and a novel spatiotemporal two-dimensional (2D) channel coding scheme based on this technology for HRLLC systems was proposed in [17]. Moreover, the achievable data rate (ADR) performance is inherently constrained by hardware impairments (HWIs), including phase-shift quantization errors and radio-frequency (RF) non-idealities [18], [19], [20]. While HWIs introduce an irreducible performance floor in the high signal-to-interference-plus-noise ratio (SINR) regime, the acquisition of accurate environment-aware CSI constitutes a distinct and fundamental bottleneck that critically governs the performance of HRLLC systems.

While massive MIMO enhances SE, the accuracy of pilot-based channel estimation is inherently limited by pilot contamination in multi-user networks [15]. Furthermore, most existing performance analyses adopt a block-fading model that assumes channel invariance within a transmission block. However, practical 6G propagation environments exhibit continuous temporal variation due to user equipment (UE) mobility and environmental dynamics, giving rise to the channel aging effect [21]. Neglecting this phenomenon leads to overly optimistic performance evaluations for HRLLC systems. Exploiting temporal correlation and developing compensatory solutions for aging channels are therefore essential [22]. The joint design of pilot spacing and block length constitutes a critical resource allocation trade-off between CSI acquisition and ADR in the presence of channel aging [23]. The impact of channel aging on the downlink performance of massive MIMO systems was investigated in [24], and its effect on cell-free (CF) massive MIMO systems under pilot contamination was further analyzed in [25]. Enhanced channel estimation precision can be achieved by incorporating more accurate statistical CSI; however, a prevalent limitation in existing literature is the assumption of perfect statistical CSI in system design and analysis. Obtaining reliable environment-aware CSI remains particularly challenging in mobility scenarios [26].

New channel characteristics have been observed in emerging 6G communication scenarios and expanded frequency bands [27], [28]. To address the challenge of acquiring accurate statistical CSI in dynamic propagation environments, the concept of the channel map has emerged as a promising solution [29], [32]. By discretizing the coverage area into quasi-stationary spatial grids, the channel map enables environment-aware statistical characterization that captures critical large-scale fading effects [30], [31]. These environment-informed priors enable precise calibration of channel estimators, ultimately enhancing the operational robustness and reliability of the system [33].

Motivated by the above challenges, this paper proposes a channel map-based CSI acquisition framework that exploits environmental information to enhance statistical CSI accuracy, and establishes a generalized performance analysis framework for HRLLC systems. Based on this framework, the fundamental trade-off between ADR and reliability under different block lengths is rigorously characterized. The main contributions of this paper are summarized as follows.

- A channel map-based statistical CSI acquisition method is proposed, which exploits environmental knowledge (including UE positions, base station (BS) locations, and multipath characteristics) to enhance the accuracy of statistical CSI.
- A unified analytical framework for HRLLC systems is developed to characterize the combined effects of pilot contamination, spatial correlation, channel aging, and channel map fidelity. Within this framework, closed-form expressions are derived for the normalized mean square error (NMSE) of the minimum mean square error (MMSE) channel estimation and for lower bounds on the conditional uplink SINR under maximum ratio (MR) combining. The analytical expressions are validated via Monte Carlo simulations.
- A comprehensive performance evaluation is conducted to compare channel map-assisted HRLLC systems with conventional systems lacking channel maps. The effects of UE mobility and channel map fidelity on system performance are systematically investigated. The analysis rigorously establishes a fundamental trade-off between reliability, ADR under latency constraints. Furthermore, the trade-off performance can be substantially improved through transmission block length optimization.

The remainder of this paper is organized as follows. Section II presents the channel map-based system model. Section III derives closed-form expressions for the NMSE of MMSE channel estimation. Section IV establishes the closed-form conditional SINR expression and characterizes the reliability-ADR trade-off. Section V presents simulation results and analysis. Conclusions are drawn in Section VI.

Notations. Vectors, matrices, and sets are denoted by bold lowercase letters $\mathbf{x}$, bold uppercase letters $\mathbf{X}$, and calligraphic uppercase letters $\mathcal{A}$, respectively. The $N \times N$ identity matrix is denoted by $\mathbf{I}_N$. The absolute value, Euclidean norm, and Frobenius norm are denoted by $|\cdot|$, $\|\cdot\|$, and $\|\cdot\|_F$, respectively. The inverse and conjugate transpose of a matrix $\mathbf{X}$ are denoted by $\mathbf{X}^{-1}$ and $\mathbf{X}^H$, respectively. The relation $\mathbf{X} \succeq \mathbf{0}$ means that $\mathbf{X}$ is positive semidefinite. The operators $\mathbb{E}\{\cdot\}$, $\mathbb{P}\{\cdot\}$, $\text{Cov}\{\cdot, \cdot\}$, and $\text{tr}(\cdot)$ denote the expectation, probability, cross-covariance, and trace, respectively. When $\text{Cov}\{\cdot\}$ is applied to a single argument, it reduces to the auto-covariance. The sets of N -dimensional real vectors, N -dimensional complex vectors, and $N \times M$ complex matrices are denoted by $\mathbb{R}^N$, $\mathbb{C}^N$, and $\mathbb{C}^{N \times M}$, respectively. The notations $\mathbf{x} \sim \mathcal{CN}(\boldsymbol{\mu}, \mathbf{R})$ and $\mathbf{y} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{R})$ denote circularly symmetric complex and real Gaussian random vectors, respectively, each with mean $\boldsymbol{\mu}$ and covariance matrix $\mathbf{R}$. The imaginary unit is denoted by j .

II. CHANNEL MAP-BASED SYSTEM MODEL

We consider a multi-user cellular MIMO network consisting of L BSs and K UEs. All BSs are equipped with a uniform linear array (ULA) consisting of N omnidirectional antennas. The spacing between antennas is set as half of the wavelength. Each BS operates individually in signal processing. There are K single-antenna UEs randomly located within the network area. The three-dimensional (3D) location vectors of the l th BS and the i th UE at time instant t are represented by $\mathbf{c}_l^{\text{BS}} \in \mathbb{R}^3$

TABLE I

DEFINITIONS OF SIGNIFICANT PARAMETERS IN THE SYSTEM MODEL

Symbol	Physical Meaning / Definition
L, K, N	Number of BSs, UEs, and antennas at each BS
$\tau_c, \tau_p, T_{\text{sym}}$	Block length, pilot length, and symbol duration
$\mathbf{c}_l^{\text{BS}}, \mathbf{c}_i^{\text{UE}}, d_{li}$	3D locations of the l th BS and i th UE, and their propagation distance (in meters)
$\mathbf{h}_{li,t}$	CTF between the i th UE and the l th BS at time t
$\mathbf{h}_{li,t}^{\text{LP}}, \alpha_{li}^{\text{LP}}$	LoS path component and its LSF amplitude coefficient
$\mathbf{h}_{li,t}^{\text{MP}}, \alpha_{li}^{\text{MP}}, \beta_{li}^{\text{MP}}$	Aggregate NLoS multipath component, its LSF amplitude coefficient, and its LSF power coefficient
$\gamma_{li}, P_{li}^{\text{L}}$	LoS indicator and its occurrence probability
K_{li}^{R}	Distance-determined Rician factor
$\chi_{li}^{\text{L}}, \chi_{li}^{\text{N}}$	Path losses in the LoS and NLoS propagation states
$F_{li}^{\text{L}}, F_{li}^{\text{N}}$	Shadowing terms in the LoS and NLoS propagation states
$\sigma_{\text{L}}^2, \sigma_{\text{N}}^2$	Variances of LoS and NLoS shadowing terms
$\mathcal{I}_{li}^{\text{MP}}, M_{\text{path}}$	Site-specific multipath parameter set and the number of resolvable NLoS paths
$\tilde{P}_{li}^r, \lambda_p$	Unnormalized Rayleigh-fading path power and its exponential distribution parameter
$P_{li}^r, \theta_{li}^r, \nu_{li}^r$	Normalized power, AoA, and Doppler shift of the r th NLoS path
$\bar{\theta}_{li}, \kappa_{\theta}$	Mean AoA and concentration parameter of the von Mises AoA distribution
$v_{li}, \varphi_{li}, f_c, c$	UE speed, moving direction relative to the l th BS, carrier frequency, and speed of light
$\mathbf{R}_{li}, \rho_{li,t_0t}$	Site-specific SCCF and TACF
$\mathbf{f}_{li,t}$	Innovation component of channel aging
$\hat{P}_{li}^r, \hat{\theta}_{li}^r, \hat{\nu}_{li}^r$	Channel map-provided normalized power, AoA, and Doppler shift of the r th path
$\hat{\gamma}_{li}^{\text{CM}}, \hat{\mathbf{R}}_{li}^{\text{CM}}, \hat{\rho}_{li,t_0t}^{\text{CM}}$	Channel map-provided LoS indicator, SCCF, and TACF
$\delta_{\gamma,li}$	Binary error in the channel map-provided LoS indicator
$\delta_{P,li}^r, \delta_{\theta,li}^r, \delta_{\nu,li}^r$	Residual errors of the log-power, AoA, and Doppler shift of the r th path
$\sigma_{\delta,P}^2, \sigma_{\delta,\theta}^2, \sigma_{\delta,\nu}^2$	Variances of the residual errors $\delta_{P,li}^r, \delta_{\theta,li}^r, \delta_{\nu,li}^r$
$\varepsilon_{\gamma}, \varepsilon_{\rho}, \varepsilon_{\mathbf{R}}$	Error levels of channel map-provided LoS indicator, TACF, and SCCF
Π_{li}, Φ_{li}	Posterior LoS probability and its second-order statistic
A_{li}, B_{li}	Mean NLoS multipath power coefficients under LoS and NLoS propagation states

and $\mathbf{c}_i^{\text{UE}} \in \mathbb{R}^3$, respectively. The propagation distance between the i th UE and the l th BS is denoted by $d_{li} = \|\mathbf{c}_i^{\text{UE}} - \mathbf{c}_l^{\text{BS}}\|$.

The system operates in time division duplex (TDD) mode [34]. Specifically, each transmission block occupies τ_c symbols, where τ_p and $\tau_c - \tau_p \geq 1$ symbols are utilized for pilot training and data transmission, respectively. Additionally, time instant corresponding to the η th symbol within a given block is expressed as $t = (\eta - 1)T_{\text{sym}}$ where T_{sym} means symbol duration.

A. Channel Model

Based on the 6G pervasive channel model in [35], channel transfer function (CTF) $\mathbf{h}_{li,t}$ between the l th BS and the i th UE at time instant t is represented by

$$\mathbf{h}_{li,t} = 10^{\frac{-\chi_{li}^{\text{L}} - F_{li}^{\text{L}}}{20}} \left(\sqrt{\frac{K_{li}^{\text{R}}}{K_{li}^{\text{R}} + 1}} \mathbf{h}_{li,t}^{\text{LP}} + \sqrt{\frac{1}{K_{li}^{\text{R}} + 1}} \mathbf{h}_{li,t}^{\text{MP}} \right) \gamma_{li} + 10^{\frac{-\chi_{li}^{\text{N}} - F_{li}^{\text{N}}}{20}} \mathbf{h}_{li,t}^{\text{MP}} (1 - \gamma_{li}) \quad (1)$$

where $K_{li}^{\text{R}} = 10^{1.3 - 0.003d_{li}}$ denotes the distance-determined Rician factor and $\gamma_{li} \sim \text{Bernoulli}(P_{li}^{\text{L}})$ represents the line-of-sight (LoS) indicator with the probability P_{li}^{L} , where $\gamma_{li} = 1$

indicates the presence of the LoS component, whereas $\gamma_{li} = 0$ indicates its absence. In addition, the path losses in the LoS and non-line-of-sight (NLoS) propagation states are denoted by χ_{li}^{L} and χ_{li}^{N} , respectively. The shadowing terms in the LoS and NLoS propagation states are $F_{li}^{\text{L}} \sim \mathcal{N}(0, \sigma_{\text{L}}^2)$ and $F_{li}^{\text{N}} \sim \mathcal{N}(0, \sigma_{\text{N}}^2)$, respectively. The vector $\mathbf{h}_{li,t}^{\text{LP}}$ denotes the deterministic LoS path component associated with the direct propagation path. In addition, $\mathbf{h}_{li,t}^{\text{MP}}$ represents the aggregate NLoS multipath component, which arises from non-direct propagation mechanisms such as reflection, scattering, and diffraction. Note that the aggregate NLoS multipath component coexists with the LoS component under $\gamma_{li} = 1$. To simplify the notation, we define the large-scale fading (LSF) amplitude coefficients of $\mathbf{h}_{li,t}^{\text{LP}}$ and $\mathbf{h}_{li,t}^{\text{MP}}$ as

$$\alpha_{li}^{\text{LP}} \triangleq \gamma_{li} 10^{\frac{-\chi_{li}^{\text{L}} - F_{li}^{\text{L}}}{20}} \sqrt{\frac{K_{li}^{\text{R}}}{K_{li}^{\text{R}} + 1}} \quad (2)$$

$$\alpha_{li}^{\text{MP}} \triangleq \gamma_{li} 10^{\frac{-\chi_{li}^{\text{L}} - F_{li}^{\text{L}}}{20}} \sqrt{\frac{1}{K_{li}^{\text{R}} + 1}} + (1 - \gamma_{li}) 10^{\frac{-\chi_{li}^{\text{N}} - F_{li}^{\text{N}}}{20}}. \quad (3)$$

Therefore, (1) can be written as

$$\mathbf{h}_{li,t} = \alpha_{li}^{\text{LP}} \mathbf{h}_{li,t}^{\text{LP}} + \alpha_{li}^{\text{MP}} \mathbf{h}_{li,t}^{\text{MP}}. \quad (4)$$

Here, the LSF coefficients are treated as time-independent, since their variation over the relevant spatial scales is negligible. This approximation is justified because both the antenna spacing (< 0.1 m) and the UE moving distance (< 0.1 m) within a single short block are far smaller than the total BS-to-UE propagation distance (e.g., 100 m).

For the small-scale fading coefficients, the site-specific multipath parameters of the link between the i th UE and the l th BS are collected in the set $\mathcal{I}_{li}^{\text{MP}} \triangleq \{P_{li}^r, \theta_{li}^r, \nu_{li}^r\}_{r=1}^{M_{\text{path}}}$, where M_{path} represents the number of resolvable NLoS paths, P_{li}^r , θ_{li}^r , and ν_{li}^r are the normalized power, angle of arrival (AoA), and Doppler shift of the r th NLoS path, respectively. Here, the path normalized power is computed as $P_{li}^r = \tilde{P}_{li}^r / \sum_{r'=1}^{M_{\text{path}}} \tilde{P}_{li}^{r'}$, where $\tilde{P}_{li}^r \sim \text{Exp}(\lambda_p)$ represents the unnormalized power with the parameter λ_p . The AoA follows the von Mises (VM) distribution $\theta_{li}^r \sim \text{VM}(\bar{\theta}_{li}, \kappa_{\theta})$ with mean direction $\bar{\theta}_{li}$ and concentration parameter κ_{θ} . The Doppler shift is expressed as $\nu_{li}^r = (f_c v_{li} / c) \cos(\theta_{li}^r - \varphi_{li})$, where c is the speed of light, v_{li} and φ_{li} denote the moving speed and direction of the i th UE relative to the l th BS. Given the multipath parameter set $\mathcal{I}_{li}^{\text{MP}}$, the conditional spatial cross-correlation function (SCCF) is [35]

$$\mathbf{R}_{li} \triangleq \mathbb{E} \left\{ \mathbf{h}_{li,t}^{\text{MP}} (\mathbf{h}_{li,t}^{\text{MP}})^{\text{H}} \middle| \mathcal{I}_{li}^{\text{MP}} \right\} = \sum_{r=1}^{M_{\text{path}}} P_{li}^r \mathbf{a}_N(\theta_{li}^r) \mathbf{a}_N^{\text{H}}(\theta_{li}^r) \quad (5)$$

where $\mathbf{a}_N(\theta) = [1, e^{-j\pi \cos \theta}, \dots, e^{-j\pi(N-1) \cos \theta}]^{\text{T}}$ denotes the steering vector. Similarly, given the multipath parameter set $\mathcal{I}_{li}^{\text{MP}}$, the conditional temporal autocorrelation function (TACF) is defined as [37]

$$\rho_{li,t_0t} \triangleq \mathbb{E} \left\{ [\mathbf{h}_{li,t_0}^{\text{MP}}]^* [\mathbf{h}_{li,t}^{\text{MP}}] \middle| \mathcal{I}_{li}^{\text{MP}} \right\} = \sum_{r=1}^{M_{\text{path}}} P_{li}^r e^{j2\pi(t-t_0)\nu_{li}^r}. \quad (6)$$

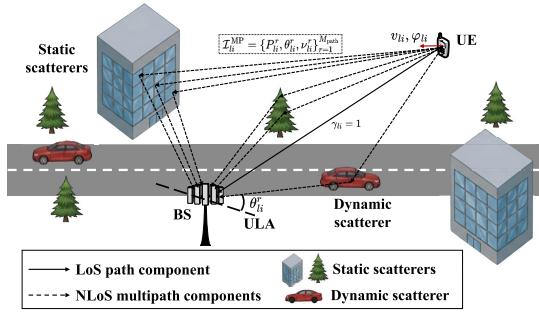


Fig. 1. Illustration of site-specific propagation multipath parameters.

Here, the expectations in (5) and (6) are taken over the random vector $\mathbf{h}_{li,t}^{MP}$. Due to $\mathbf{a}_N^H(\theta)\mathbf{a}_N(\theta) = N$ and $\sum_{r=1}^{M_{\text{path}}} P_{li}^r = 1$, we have $\text{tr}(\mathbf{R}_{li}) = N$, $\mathbf{R}_{li} \succeq \mathbf{0}$, and $0 \leq |\rho_{li,t_0t}| \leq 1$. The term “site-specific” emphasizes that these parameters in $\mathcal{I}_{li}^{MP}$ are determined by the geometry of the particular UE-BS link rather than by a universal isotropic assumption. Across different locations, they follow the distributions specified below, whereas for a given scenario they take fixed values. These fixed values yield the true second-order statistics of the aggregate NLoS multipath component $\mathbf{h}_{li,t}^{MP}$. Without loss of generality, $\mathbf{h}_{li,t}^{MP}$ is assumed to be mutually independent across different indices i and l , and the channel is assumed to be stationary.

The movement of the UEs results in the temporal variations of $\mathbf{h}_{li,t}$ across different time instants t within a single transmission block, called channel aging [24]. The temporal variations of the NLoS component $\mathbf{h}_{li,t}^{MP}$ can be modeled as [25]

$$\mathbf{h}_{li,t}^{MP} = \rho_{li,t_0t} \mathbf{h}_{li,t_0}^{MP} + \sqrt{1 - |\rho_{li,t_0t}|^2} \mathbf{f}_{li,t} \quad (7)$$

where $\mathbf{f}_{li,t} | \mathcal{I}_{li}^{MP} \sim \mathcal{CN}(\mathbf{0}, \mathbf{R}_{li})$ means the innovation component independent of initial state $\mathbf{h}_{li,t_0}^{MP}$.

B. Channel Map-Based CSI Acquisition

As illustrated in Fig. 1, the LoS path component propagates directly from the UE to the BS along an unobstructed path. In contrast, the NLoS multipath components reach the BS after interacting with surrounding static or dynamic scatterers, such as buildings, trees, and moving vehicles. Each resolvable path is characterized by its power P_{li}^r , AoA θ_{li}^r , and Doppler shift ν_{li}^r , which are collected in the site-specific parameter set $\mathcal{I}_{li}^{MP}$. From a modeling perspective, these environmental features play two distinct roles. They act either as propagation obstructions that govern the LoS indicator, or as scatterers that shape the small-scale fading parameters. This dichotomy motivates the exploitation of the environmental geometry information for statistical CSI acquisition.

In conventional massive MIMO systems, the acquisition of statistical CSI typically relies on geometry-agnostic empirical models, such as the classical Jakes spectrum for temporal correlation. These models assume isotropic or homogeneous scattering and therefore fail to capture the site-specific propagation characteristics of a given cell. To overcome this fundamental limitation, the channel map emerges as a

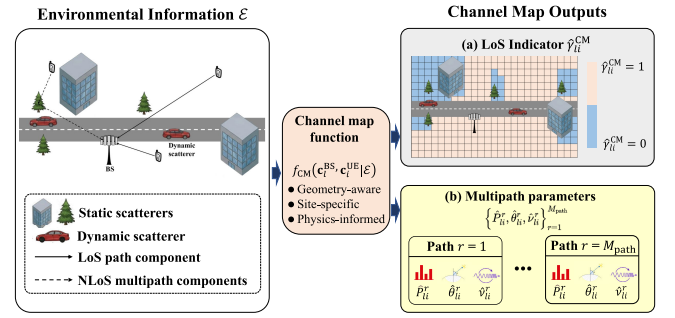


Fig. 2. Illustration of channel map-based CSI acquisition.

geometry-aware paradigm to provide statistical CSI. Specifically, the channel map is a site-specific database constructed from the environmental information, which encapsulates the BS locations $\mathbf{c}_l^{\text{BS}}$ and the geometric parameters of the physical scatterers. As illustrated in Fig. 2, the channel map infers the LoS state and the multipath parameters from the environmental information, including the BS, UE, and scatterer locations. The mapping function $f_{\text{CM}}(\cdot)$ is geometry-aware, site-specific, and physics-informed, since it exploits the deterministic geometric relationships and scattering mechanisms to capture the propagation characteristics of the particular cell. Through these properties, the channel map establishes a deterministic mapping from the physical coordinate space to the statistical CSI. Therefore, this mapping relationship is formulated as

$$\{\hat{P}_{li}^r, \hat{\nu}_{li}^r, \hat{\theta}_{li}^r, \hat{\gamma}_{li}^{CM}\} = f_{\text{CM}}(\mathbf{c}_l^{\text{BS}}, \mathbf{c}_i^{\text{UE}} | \mathcal{E}) \quad (8)$$

where $\mathcal{E}$ denotes the environmental information, including the obstruction layout, scatterer geometry, and mobility-related environmental features, $f_{\text{CM}}(\cdot)$ represents the channel map mapping function, $\hat{\gamma}_{li}^{CM}$ means the channel map-provided LoS indicator, $\hat{P}_{li}^r$, $\hat{\nu}_{li}^r$, and $\hat{\theta}_{li}^r$ are channel map-provided multipath parameters corresponding to P_{li}^r , ν_{li}^r , and θ_{li}^r , respectively.

While channel maps significantly enhance the acquisition of statistical CSI, the real-time maintenance of dynamic environmental features inevitably introduces computational and signaling overhead. In practice, the fidelity of a channel map is constrained by the limited resources available for its construction and periodic updates. Nevertheless, the systematic design of map-updating algorithms is beyond the scope of this work. To maintain analytical tractability while accounting for these practical limitations, the cost of channel map acquisition is characterized through the residual estimation error of the obtained CSI. For the LoS indicator, the statistical error model of $\hat{\gamma}_{li}^{CM}$ is modeled as

$$\hat{\gamma}_{li}^{CM} = (1 - \delta_{\gamma,li})\gamma_{li} + \delta_{\gamma,li}(1 - \gamma_{li}) \quad (9)$$

where $\delta_{\gamma,li} \sim \text{Bernoulli}(\varepsilon_\gamma)$ is a binary error with the $\mathbb{P}(\delta_{\gamma,li} = 1) = \varepsilon_\gamma$ satisfying $0 \leq \varepsilon_\gamma \leq 1/2$. The channel map-provided multipath parameters are generated from the true site-specific multipath parameters with residual estimation errors. Specifically, multipath parameters are modeled as

$$\begin{cases} \hat{P}_{li}^r = \frac{P_{li}^r \exp(\delta_{P,li}^r)}{\sum_{r'=1}^{M_{\text{path}}} P_{li}^{r'} \exp(\delta_{P,li}^{r'})} \\ \hat{\theta}_{li}^r = \theta_{li}^r + \delta_{\theta,li}^r \\ \hat{\nu}_{li}^r = \nu_{li}^r + \delta_{\nu,li}^r \end{cases} \quad (10)$$

where $\delta_{P,li}^r \sim \mathcal{N}(0, \sigma_{\delta,P}^2)$, $\delta_{\theta,li}^r \sim \mathcal{N}(0, \sigma_{\delta,\theta}^2)$, and $\delta_{\nu,li}^r \sim \mathcal{N}(0, \sigma_{\delta,\nu}^2)$ denote the residual errors of the path log-power, the AoA, and the Doppler shift, respectively. In particular, the residual error $\delta_{\nu,li}^r$ additionally stems from variations in the moving speed and direction of the UE. Based on these channel map-provided multipath parameters, the SCCF and TACF provided by channel maps are

$$\hat{\mathbf{R}}_{li}^{\text{CM}} = \sum_{r=1}^{M_{\text{path}}} \hat{P}_{li}^r \mathbf{a}_N(\hat{\theta}_{li}^r) \mathbf{a}_N^H(\hat{\theta}_{li}^r) \quad (11)$$

$$\hat{\rho}_{li,tot}^{\text{CM}} = \sum_{r=1}^{M_{\text{path}}} \hat{P}_{li}^r e^{j2\pi(t-t_0)\hat{\nu}_{li}^r}. \quad (12)$$

Based on (9)–(12), these channel map-provided prior information satisfy $\hat{\gamma}_{li}^{\text{CM}} \in \{0, 1\}$, $\hat{\mathbf{R}}_{li}^{\text{CM}} \succeq \mathbf{0}$, and $0 \leq |\hat{\rho}_{li,tot}^{\text{CM}}| \leq 1$.

Introducing multipath parameters grounded in the actual propagation environment improves the physical interpretability of the error source. The channel map error levels are jointly controlled by adjusting the perturbation strengths $\sigma_{\delta,P}^2$, $\sigma_{\delta,\theta}^2$, and $\sigma_{\delta,\nu}^2$ of the multipath parameters. The resulting channel map fidelity is quantified by

$$\varepsilon_{\mathbf{R}}^2 \triangleq \frac{\mathbb{E} \left\{ \left\| \hat{\mathbf{R}}_{li}^{\text{CM}} - \mathbf{R}_{li} \right\|_{\text{F}}^2 \right\}}{\mathbb{E} \left\{ \left\| \mathbf{R}_{li} \right\|_{\text{F}}^2 \right\}}, \quad \varepsilon_{\rho}^2 \triangleq \frac{\mathbb{E} \left\{ \left| \hat{\rho}_{li,tot}^{\text{CM}} - \rho_{li,tot} \right|^2 \right\}}{\mathbb{E} \left\{ \left| \rho_{li,tot} \right|^2 \right\}}. \quad (13)$$

Here, the expectations in (13) are taken over the perturbations $\delta_{P,li}^r$, $\delta_{\theta,li}^r$, $\delta_{\nu,li}^r$ and the site-specific parameters in $\mathcal{I}_{li}^{\text{MP}}$. For different $\varepsilon_{\mathbf{R}}$ and ε_{ρ} , the parameters $\sigma_{\delta,P}^2$, $\sigma_{\delta,\theta}^2$, and $\sigma_{\delta,\nu}^2$ are calibrated offline such that $\hat{\mathbf{R}}_{li}^{\text{CM}}$ and $\hat{\rho}_{li,tot}^{\text{CM}}$ satisfy (13).

To simplify the notation, we denote the LSF power coefficient of aggregate NLoS multipath component by $\beta_{li}^{\text{MP}} \triangleq (\alpha_{li}^{\text{MP}})^2$. Based on (9), the channel map-based estimates of the required LSF coefficients are given by

$$\hat{\alpha}_{li}^{\text{LP,CM}} = \Pi_{li} \sqrt{\frac{K_{li}^{\text{R}}}{K_{li}^{\text{R}} + 1}} \exp \left(-\frac{\xi_{\text{dB}}}{2} \chi_{li}^{\text{L}} + \frac{\xi_{\text{dB}}^2}{8} \sigma_{\text{L}}^2 \right) \quad (14)$$

$$\hat{\beta}_{li}^{\text{MP,CM}} = \Pi_{li} A_{li} + (1 - \Pi_{li}) B_{li} \quad (15)$$

where $\xi_{\text{dB}} = \ln(10)/10$ denotes the dB conversion factor, power coefficients A_{li} and B_{li} are expressed as

$$A_{li} \triangleq \frac{1}{K_{li}^{\text{R}} + 1} \exp \left(-\xi_{\text{dB}} \chi_{li}^{\text{L}} + \frac{\xi_{\text{dB}}^2}{2} \sigma_{\text{L}}^2 \right) \quad (16)$$

$$B_{li} \triangleq \exp \left(-\xi_{\text{dB}} \chi_{li}^{\text{N}} + \frac{\xi_{\text{dB}}^2}{2} \sigma_{\text{N}}^2 \right). \quad (17)$$

Here, the posterior probability Π_{li} is given by

$$\begin{aligned} \Pi_{li} &\triangleq \mathbb{P}(\gamma_{li} = 1 \mid \hat{\gamma}_{li}^{\text{CM}}) \\ &= \left[1 + \frac{1 - P_{li}^{\text{L}}}{P_{li}^{\text{L}}} \left(\frac{\varepsilon_{\gamma}}{1 - \varepsilon_{\gamma}} \right)^{2\hat{\gamma}_{li}^{\text{CM}} - 1} \right]^{-1}. \end{aligned} \quad (18)$$

Theorem 1: With the CSI provided from channel maps, we can obtain NMSE as

$$\text{NMSE}(\alpha_{li}^{\text{LP}}, \hat{\alpha}_{li}^{\text{LP,CM}}) = 1 - \frac{\Phi_{li}}{P_{li}^{\text{L}}} \exp \left(-\frac{\xi_{\text{dB}}^2 \sigma_{\text{L}}^2}{4} \right) \quad (19)$$

$$\text{NMSE}(\beta_{li}^{\text{MP}}, \hat{\beta}_{li}^{\text{MP,CM}})$$

$$= 1 - \frac{\Phi_{li} (A_{li} - B_{li})^2 + 2P_{li}^{\text{L}} B_{li} (A_{li} - B_{li}) + B_{li}^2}{P_{li}^{\text{L}} A_{li}^2 e^{\xi_{\text{dB}}^2 \sigma_{\text{L}}^2} + (1 - P_{li}^{\text{L}}) B_{li}^2 e^{\xi_{\text{dB}}^2 \sigma_{\text{N}}^2}} \quad (20)$$

where the NMSE is defined as $\text{NMSE}(x, \hat{x}) \triangleq \frac{\mathbb{E}\{|x - \hat{x}|^2\}}{\mathbb{E}\{|x|^2\}}$ and the second-order statistic $\Phi_{li} \triangleq \mathbb{E}\{\Pi_{li}^2\}$ can be expressed as

$$\begin{aligned} \Phi_{li} &= \frac{(P_{li}^{\text{L}})^2 (1 - \varepsilon_{\gamma})^2}{P_{li}^{\text{L}} (1 - \varepsilon_{\gamma}) + (1 - P_{li}^{\text{L}}) \varepsilon_{\gamma}} \\ &\quad + \frac{(P_{li}^{\text{L}})^2 \varepsilon_{\gamma}^2}{P_{li}^{\text{L}} \varepsilon_{\gamma} + (1 - P_{li}^{\text{L}}) (1 - \varepsilon_{\gamma})}. \end{aligned} \quad (21)$$

Proof: Please refer to Appendix A.

It is worth examining the asymptotic behavior of the LoS detection error probability ε_{γ} . Since Π_{li} is a posterior probability, we have $0 \leq \Pi_{li} \leq 1$ and $\mathbb{E}\{\Pi_{li}\} = P_{li}^{\text{L}}$, where the expectation is taken over random variables (RVs) γ_{li} and $\delta_{\gamma,li}$. Furthermore, its second-order statistic Φ_{li} satisfies $(P_{li}^{\text{L}})^2 \leq \Phi_{li} \leq P_{li}^{\text{L}}$, where the lower bound follows from Jensen's inequality and the upper bound follows from $\Pi_{li}^2 \leq \Pi_{li}$. This bounding property characterizes the asymptotic behavior of the proposed channel map-assisted estimators at the two endpoints of the informative map-reliability regime.

- **Perfect Channel Map** ($\varepsilon_{\gamma} \rightarrow 0$): In this case, the LoS state is perfectly detected, and the posterior probability degenerates into a deterministic binary indicator. Thus, $\Pi_{li} \rightarrow \gamma_{li}$ and $\Phi_{li} \rightarrow P_{li}^{\text{L}}$. The NMSEs converge to

$$\text{NMSE}(\alpha_{li}^{\text{LP}}, \hat{\alpha}_{li}^{\text{LP,CM}}) \rightarrow 1 - \exp \left(-\frac{\xi_{\text{dB}}^2 \sigma_{\text{L}}^2}{4} \right) \quad (22)$$

$$\begin{aligned} &\text{NMSE}(\beta_{li}^{\text{MP}}, \hat{\beta}_{li}^{\text{MP,CM}}) \\ &\rightarrow 1 - \frac{P_{li}^{\text{L}} A_{li}^2 + (1 - P_{li}^{\text{L}}) B_{li}^2}{P_{li}^{\text{L}} A_{li}^2 e^{\xi_{\text{dB}}^2 \sigma_{\text{L}}^2} + (1 - P_{li}^{\text{L}}) B_{li}^2 e^{\xi_{\text{dB}}^2 \sigma_{\text{N}}^2}}. \end{aligned} \quad (23)$$

These results show that the estimation errors do not vanish even with a perfect channel map. The residual error floor is caused by the stochastic log-normal shadowing, which is not fully specified by the LoS indicator.

- **Uninformative Boundary** ($\varepsilon_{\gamma} \rightarrow 1/2$). The value $\varepsilon_{\gamma} \rightarrow 1/2$ marks the boundary at which the LoS indicator provided by the channel map becomes uninformative. At this boundary, the indicator is statistically independent of the true LoS state. Consequently, the channel map-based LSF estimator reduces to the conventional LSF estimator based solely on the LoS probability P_{li}^{L} . When $\varepsilon_{\gamma} > 1/2$, the indicator becomes negatively correlated with the true LoS state, and the resulting estimator performs worse than the conventional one.

C. CSI Acquisition Without Channel Maps

In the absence of site-specific environmental information, the BS lacks deterministic knowledge of the precise locations of obstructions and scatterers. Consequently, the system must resort to standardized statistical models, commonly referred to as empirical CSI, that presuppose a generalized, geometry-agnostic scattering environment.

For the LSF coefficients, the estimation degenerates to an unconditional statistical expectation based solely on the

distance-dependent prior probability P_{li}^L . By replacing the map-assisted posterior probability Π_{li} with the prior probability P_{li}^L , the prior-based estimates are given by

$$\hat{\alpha}_{li}^{\text{LP,nCM}} = P_{li}^L \sqrt{\frac{K_{li}^R}{K_{li}^R + 1}} \exp\left(-\frac{\xi_{\text{dB}}}{2} \chi_{li}^L + \frac{\xi_{\text{dB}}^2}{8} \sigma_L^2\right) \quad (24)$$

$$\hat{\beta}_{li}^{\text{MP,nCM}} = P_{li}^L A_{li} + (1 - P_{li}^L) B_{li}. \quad (25)$$

As shown in (25), the prior-based estimate coincides with the map-assisted estimate $\hat{\beta}_{li}^{\text{MP,CM}}$ at the uninformative boundary $\varepsilon_\gamma \rightarrow 1/2$, where $\Pi_{li} \rightarrow P_{li}^L$.

Regarding the small-scale fading parameters, the absence of geometric data necessitates generalized stochastic assumptions for both spatial and temporal correlations. In the spatial domain, lacking the site-specific angular parameter $\bar{\theta}_{li}$, the BS adopts the isotropic assumption $\kappa_\theta \rightarrow 0$, under which the von Mises distribution reduces to the uniform distribution over $(0, 2\pi)$. Therefore, the SCCF and TACF are [21], [30]

$$\begin{aligned} [\hat{\mathbf{R}}_{li}^{\text{nCM}}]_{n,\tilde{n}} &= \sum_{r=1}^{M_{\text{path}}} \mathbb{E}\left\{P_{li}^r e^{-j\pi(n-\tilde{n})\cos(\theta_{li}^r)}\right\} \\ &= J_0(\pi(n-\tilde{n})) \end{aligned} \quad (26)$$

$$\begin{aligned} \hat{\rho}_{li,t_0t}^{\text{nCM}} &= \sum_{r=1}^{M_{\text{path}}} \mathbb{E}\left\{P_{li}^r e^{j2\pi(t-t_0)\nu_{li}^r}\right\} \\ &= J_0(2\pi f_c \nu_{li}(t-t_0)/c) \end{aligned} \quad (27)$$

where $J_0(\cdot)$ represents the zeroth-order Bessel function of the first kind. Here, the expectations in (26) and (27) are taken over the site-specific parameters in $\mathcal{I}_{li}^{\text{MP}}$. Under stationary channels, M_{path} is usually set to a scenario-dependent fixed value, consistent with the commonly used channel models [35], [36], [37]. This leads naturally to the classical Jakes spectrum.

III. PILOT TRANSMISSION WITH IMPERFECT STATISTICAL CSI

In this section, a novel closed-form expression of NMSE of MMSE channel estimator is derived, considering imperfect statistical CSI, channel aging effect, channel spatial correlation, and pilot contamination. Subsequently, simplified cases applicable to distinct scenarios are given.

During the pilot phase, τ_p mutually orthogonal pilot sequences are available and can be reused among UEs. Each UE transmits one pilot sequence for channel estimation. Since $\tau_p \ll K$, pilot sequences may be reused by different UEs, which causes the pilot contamination effect. Let $\eta_i \in \{1, \dots, \tau_p\}$ denote the pilot index assigned to the i th UE and $t_i^P = (\eta_i - 1)T_{\text{sym}}$ represent the corresponding pilot time instant. We denote the set of UEs sharing the η_k th pilot by $\mathcal{S}_k^P = \{i \mid t_i^P = t_k^P, i = 1, \dots, K\}$. The received pilot observation at the l th BS associated with the k th UE is [25]

$$\mathbf{y}_{l,t_k^P} = \sum_{i \in \mathcal{S}_k^P} \sqrt{p_i} \mathbf{h}_{li,t_k^P} + \mathbf{z}_{l,t_k^P} \quad (28)$$

where $p_i \geq 0$ represents the transmit power of the i th UE, and $\mathbf{z}_{l,t_k^P} \sim \mathcal{CN}(\mathbf{0}, \sigma_z^2 \mathbf{I}_N)$ denotes the additive noise. The channel

estimation is completed at $t_{\tau_p} = \tau_p T_{\text{sym}}$, and the resulting estimate is used as the initial channel state for subsequent data transmission [21]. Although carrier frequency offset critically degrades conventional wideband systems [8], [9], its impact is marginal in the considered HRLLC system. The inter-carrier interference is mitigated due to the single-carrier transmission adopted in this work. In addition, the short block length renders the accumulated phase drift negligible. Therefore, perfect carrier frequency synchronization is assumed, consistent with commonly used analytical frameworks [12], [13], [14], [15].

In practical HRLLC systems, such perfect prior knowledge is unattainable. Instead, the BSs must rely on the acquired statistical and LoS indicator set, denoted as

$$\mathcal{I}_{l,S_k^P} = \{\hat{\alpha}_{li}^{\text{LP}}, \hat{\beta}_{li}^{\text{MP}}, \hat{\rho}_{li,t_k^P t_{\tau_p}}, \hat{\mathbf{R}}_{li}\}_{i \in \mathcal{S}_k^P}. \quad (29)$$

Depending on the network configuration, these parameters are acquired either via the proposed channel map or through conventional empirical models, i.e.,

$$\begin{aligned} \hat{\alpha}_{li}^{\text{LP}} &\in \{\hat{\alpha}_{li}^{\text{LP,CM}}, \hat{\alpha}_{li}^{\text{LP,nCM}}\} & \hat{\rho}_{li,t_0t} &\in \{\hat{\rho}_{li,t_0t}^{\text{CM}}, \hat{\rho}_{li,t_0t}^{\text{nCM}}\} \\ \hat{\beta}_{li}^{\text{MP}} &\in \{\hat{\beta}_{li}^{\text{MP,CM}}, \hat{\beta}_{li}^{\text{MP,nCM}}\} & \hat{\mathbf{R}}_{li} &\in \{\hat{\mathbf{R}}_{li}^{\text{CM}}, \hat{\mathbf{R}}_{li}^{\text{nCM}}\}. \end{aligned} \quad (30)$$

Here, the superscripts CM and nCM denote the channel map-based and non-channel-map cases, respectively. In addition, we define the conditioning set $\mathcal{G}_l$ as $\mathcal{G}_l \triangleq \mathcal{I}_{l,S_k^P} \cup \{\alpha_{li}^{\text{LP}}, \beta_{li}^{\text{MP}}\}_{i=1}^K$ to simplify the notation.

As a theoretical benchmark, assuming that BSs have perfect knowledge of the LSF coefficients (α_{li}^{LP} and β_{li}^{MP}) and the statistical CSI ($\rho_{li,t_k^P t_{\tau_p}}$ and $\mathbf{R}_{li}$) at the BSs, the MMSE estimate of $\mathbf{h}_{lk,t_{\tau_p}}$ is expressed as [25]

$$\begin{aligned} \hat{\mathbf{h}}_{lk,t_{\tau_p}}^{\text{Pe}} &= \alpha_{lk}^{\text{LP}} \mathbf{h}_{lk,t_{\tau_p}}^{\text{LP}} \\ &+ \sqrt{p_k} \beta_{lk}^{\text{MP}} \rho_{lk,t_k^P t_{\tau_p}} \mathbf{R}_{lk} \Psi_{l,t_k^P}^{-1} (\mathbf{y}_{l,t_k^P} - \bar{\mathbf{y}}_{l,t_k^P}) \end{aligned} \quad (31)$$

where the true covariance matrix and mean vector of the received pilot observation are given by $\Psi_{l,t_k^P} \triangleq \sigma_z^2 \mathbf{I}_N + \sum_{i \in \mathcal{S}_k^P} p_i \beta_{li}^{\text{MP}} \mathbf{R}_{li}$ and $\bar{\mathbf{y}}_{l,t_k^P} \triangleq \sum_{i \in \mathcal{S}_k^P} \sqrt{p_i} \alpha_{li}^{\text{LP}} \mathbf{h}_{li,t_k^P}^{\text{LP}}$, respectively.

According to the properties of the MMSE estimator, the estimated result $\hat{\mathbf{h}}_{lk,t_{\tau_p}}^{\text{Pe}}$ follows $\mathcal{CN}(\alpha_{lk}^{\text{LP}} \mathbf{h}_{lk,t_{\tau_p}}^{\text{LP}}, \mathbf{Q}_{lk,li,t_{\tau_p}})$, where the conditional cross-covariance matrix $\mathbf{Q}_{lk,li,t_{\tau_p}}$ is defined as

$$\begin{aligned} \mathbf{Q}_{lk,li,t_{\tau_p}} &\triangleq \text{Cov}\left\{\hat{\mathbf{h}}_{lk,t_{\tau_p}}^{\text{Pe}}, \hat{\mathbf{h}}_{li,t_{\tau_p}}^{\text{Pe}} \mid \mathcal{G}_l\right\} \\ &= \begin{cases} \sqrt{p_k p_i} \beta_{lk}^{\text{MP}} \beta_{li}^{\text{MP}} \rho_{lk,t_k^P t_{\tau_p}} \rho_{li,t_k^P t_{\tau_p}}^* \\ \times \mathbf{R}_{lk} \Psi_{l,t_k^P}^{-1} \mathbf{R}_{li}, \\ i \in \mathcal{S}_k^P \\ \mathbf{0}_{N \times N}, i \notin \mathcal{S}_k^P \end{cases} \end{aligned} \quad (32)$$

where the covariance is taken over RVs $\mathbf{h}_{li,t}^{\text{MP}}$ and $\mathbf{z}_{l,t_k^P}$.

Due to limited map fidelity or model mismatch, the acquired parameter set $\mathcal{I}_{l,S_k^P}$ is inevitably imperfect. To perform channel estimation under these practical constraints, the BS invokes the certainty-equivalence principle by directly substituting the imperfect parameter estimates into the optimal MMSE structure in (31). The resulting mismatched estimator, whose

performance is degraded by statistical parameter mismatches, is given as follows.

Lemma 1: Given the set $\mathcal{G}_l$, the practical estimate based on the equivalent MMSE estimator can be expressed as

$$\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}} = \hat{\alpha}_{lk}^{\text{LP}} \mathbf{h}_{lk,t\tau_p}^{\text{LP}} + \sqrt{p_k} \hat{\beta}_{lk}^{\text{MP}} \hat{\rho}_{lk,t_k^P t_{\tau_p}} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} (\mathbf{y}_{l,t_k^P} - \hat{\mathbf{y}}_{l,t_k^P}) \quad (33)$$

where the estimated covariance matrix and mean vector are given by $\hat{\Psi}_{l,t_k^P} \triangleq \sum_{i \in \mathcal{S}_k^P} p_i \hat{\beta}_{li}^{\text{MP}} \hat{\mathbf{R}}_{li} + \sigma_z^2 \mathbf{I}_N$ and $\hat{\mathbf{y}}_{l,t_k^P} \triangleq \sum_{i \in \mathcal{S}_k^P} \sqrt{p_i} \hat{\alpha}_{li}^{\text{LP}} \mathbf{h}_{li,t_k^P}^{\text{LP}}$, respectively.

Proof: The estimator in (33) is obtained by directly substituting the acquired parameter set $\mathcal{I}_{l,S_k^P}$ into the MMSE criterion detailed in [34, Lemma B.19].

Furthermore, conditioned on the set $\mathcal{G}_l$, the practical estimate $\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}}$ follows a complex Gaussian distribution. Its conditional mean and covariance matrix are

$$\mathbb{E}\{\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}} | \mathcal{G}_l\} = \hat{\alpha}_{lk}^{\text{LP}} \mathbf{h}_{lk,t\tau_p}^{\text{LP}} + \mathbf{b}_{lk,t_k^P t_{\tau_p}} \quad (34)$$

$$\begin{aligned} \hat{\mathbf{Q}}_{lk,t\tau_p} &\triangleq \text{Cov}\{\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}} | \mathcal{G}_l\} \\ &= p_k \left(\hat{\beta}_{lk}^{\text{MP}} \right)^2 \left| \hat{\rho}_{lk,t_k^P t_{\tau_p}} \right|^2 \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} \hat{\Psi}_{l,t_k^P} \hat{\mathbf{R}}_{lk} \end{aligned} \quad (35)$$

where the expectation and covariance are taken over RVs $\mathbf{h}_{li,t}^{\text{MP}}$ and $\mathbf{z}_{l,t_k^P}$, the bias term $\mathbf{b}_{lk,t_k^P t_{\tau_p}}$ is expressed as

$$\mathbf{b}_{lk,t_k^P t_{\tau_p}} = \sqrt{p_k} \hat{\beta}_{lk}^{\text{MP}} \hat{\rho}_{lk,t_k^P t_{\tau_p}} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} (\bar{\mathbf{y}}_{l,t_k^P} - \hat{\mathbf{y}}_{l,t_k^P}). \quad (36)$$

Building upon these conditional statistics, we can derive the closed-form expression for the exact NMSE of the mismatched MMSE estimator.

Theorem 2: Given the set $\mathcal{G}_l$, the NMSE of the practical estimator $\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}}$ is

$$\begin{aligned} \text{NMSE}(\mathbf{h}_{lk,t\tau_p}, \hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}} | \mathcal{G}_l) &= \frac{1}{N \left[|\hat{\alpha}_{lk}^{\text{LP}}|^2 + \hat{\beta}_{lk}^{\text{MP}} \right]} \left\{ N \hat{\beta}_{lk}^{\text{MP}} \right. \\ &\quad - \text{tr}(\hat{\mathbf{Q}}_{lk,t\tau_p}) + \left\| (\hat{\alpha}_{lk}^{\text{LP}} - \alpha_{lk}^{\text{LP}}) \mathbf{h}_{lk,t\tau_p}^{\text{LP}} + \mathbf{b}_{lk,t_k^P t_{\tau_p}} \right\|^2 \\ &\quad \left. + p_k \text{tr}(\hat{\Delta}_{lk,t_k^P t_{\tau_p}} \hat{\Psi}_{l,t_k^P} \hat{\Delta}_{lk,t_k^P t_{\tau_p}}^H) \right\} \end{aligned} \quad (37)$$

where the mismatch matrix $\hat{\Delta}_{lk,t_k^P t_{\tau_p}}$ is given by

$$\hat{\Delta}_{lk,t_k^P t_{\tau_p}} = \hat{\beta}_{lk}^{\text{MP}} \hat{\rho}_{lk,t_k^P t_{\tau_p}} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} - \hat{\beta}_{lk}^{\text{MP}} \hat{\rho}_{lk,t_k^P t_{\tau_p}} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} \hat{\Psi}_{l,t_k^P} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k^P}^{-1} \quad (38)$$

Proof: Please refer to Appendix B.

Theorem 2 reveals that the NMSE of the MMSE channel estimator is governed by pilot contamination, the accuracy of statistical CSI [35], channel spatial correlation [34], and the channel aging effect [25]. Furthermore, under the condition of perfect statistical CSI, the channel estimate and its associated error are statistically uncorrelated by the orthogonality property of the MMSE estimator [34]. However, the availability of only imperfect statistical CSI may compromise this inherent orthogonality, introducing statistical dependence between the estimate and the estimation error.

IV. UPLINK DATA TRANSMISSION WITH IMPERFECT STATISTICAL CSI

In this section, a novel closed-form expression of lower bound on the uplink SINR is derived. The imperfect statistical CSI, channel aging effects, channel spatial correlation, and multi-user interference are comprehensively considered in the performance analysis. Furthermore, the trade-off relationship between the coding error rate, latency, and ADRs is revealed.

A. SINR Under Imperfect Statistical CSI

We consider the uplink data transmission phase of a multi-user massive MIMO cellular network, where each BS employs the received signals and MR combining to detect data based on its locally available mismatched channel estimates. During the uplink data phase, the received signal $\mathbf{y}_{l,t}^{\text{U}}$ at the l th BS at time instant t is given by

$$\mathbf{y}_{l,t}^{\text{U}} = \sum_{i=1}^K \sqrt{p_i} \mathbf{h}_{li,t} s_i(t) + \mathbf{z}_{l,t} \quad (39)$$

where $\mathbf{z}_{l,t} \sim \mathcal{CN}(\mathbf{0}, \sigma_z^2 \mathbf{I}_N)$ denotes the additive noise vector, $s_i(t)$ is the uplink data symbol satisfying $\mathbb{E}\{|s_i(t)|^2\} = 1$, and p_i denotes the transmit power of the i th UE.

Here, the MR combining based on the practical channel estimates is considered. To derive a tractable SINR expression, the multiple sources of interference are systematically decoupled. By invoking the use-and-then-forget bounding technique, the detected signal $\hat{s}_{lk}(t)$ is decomposed into

$$\begin{aligned} \hat{s}_{lk}(t) &\triangleq (\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}})^H \mathbf{y}_{l,t}^{\text{U}} = \underbrace{(\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Im}} - \hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{y}_{l,t}^{\text{U}}}_{\text{ISC}_{lk}(t)} \\ &\quad + \underbrace{\sqrt{p_k} \mathbb{E}\{(\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{h}_{lk,t} | \mathcal{G}_l\} s_k(t)}_{\text{DS}_{lk}(t)} \\ &\quad + \underbrace{\sqrt{p_k} \left((\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{h}_{lk,t} - \mathbb{E}\{(\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{h}_{lk,t} | \mathcal{G}_l\} \right) s_k(t)}_{\text{CGU}_{lk}(t)} \\ &\quad + \sum_{i \neq k} \underbrace{\sqrt{p_i} (\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{h}_{li,t} s_i(t)}_{\text{MUI}_{lk}(t)} + \underbrace{(\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{z}_{l,t}}_{\text{AN}_{lk}(t)} \end{aligned} \quad (40)$$

where the expectation and covariance are taken over RVs $\mathbf{h}_{li,t}^{\text{MP}}$ and $\mathbf{z}_{l,t}$. Here, $\text{ISC}_{lk}(t)$ represents the interference penalty induced by imperfect statistical CSI, $\text{DS}_{lk}(t)$ represents the desired signal, $\text{CGU}_{lk}(t)$, $\text{MUI}_{lk}(t)$, and $\text{AN}_{lk}(t)$ are interference caused by combining gain uncertainty, multi-user interference, and additive noise, respectively. Note that the channel aging effect is embedded in the temporal evolution of the inner product between the channel estimate at t_{τ_p} and the actual channel vector at t , i.e., $(\hat{\mathbf{h}}_{lk,t\tau_p}^{\text{Pe}})^H \mathbf{h}_{lk,t}$. In addition, the channel aging effect attenuates the effective coherent signal gain in the DS term through the TACF $\hat{\rho}_{lk,t\tau_p t}$, while simultaneously inflating the variance of the effective noise within the $\text{CGU}_{lk}(t)$ term.

Theorem 3: Given the imperfect statistical CSI set $\mathcal{I}_{l,S_k^P}$, a closed-form lower bound on the conditional uplink SINR of

the k th UE at the l th BS under the MR combining scheme is given by

$$\begin{aligned} \text{SINR}_{lk}(t) | \mathcal{G}_l = p_k & |\Xi_{lk, lk, t_{\tau_p} t}|^2 \\ & \cdot \left\{ \sigma_z^2 \left[N |\alpha_{lk}^{\text{LP}}|^2 + \text{tr}(\mathbf{Q}_{lk, lk, t_{\tau_p} t}) \right] + \sum_{i=1}^K p_i \Xi_{lk, li, t_{\tau_p} t}^{\text{Sq}} \right. \\ & \left. - p_k |\Xi_{lk, lk, t_{\tau_p} t}|^2 + \mathbb{E} \left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\} \right\}^{-1} \end{aligned} \quad (41)$$

where $\Xi_{lk, li, t_{\tau_p} t}$, $\mathbb{E} \left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\}$, and $\Xi_{lk, li, t_{\tau_p} t}^{\text{Sq}}$ are respectively given by

$$\begin{aligned} \Xi_{lk, li, t_{\tau_p} t} & \triangleq \mathbb{E} \left\{ \left(\hat{\mathbf{h}}_{lk, t_{\tau_p}}^{\text{Pe}} \right)^H \mathbf{h}_{li, t} | \mathcal{G}_l \right\} \\ & = \alpha_{lk}^{\text{LP}} \alpha_{li}^{\text{LP}} \left(\mathbf{h}_{lk, t_{\tau_p}}^{\text{LP}} \right)^H \mathbf{h}_{li, t}^{\text{LP}} + \begin{cases} \rho_{li, t_{\tau_p} t} \text{tr}(\mathbf{Q}_{lk, li, t_{\tau_p} t}), & i \in \mathcal{S}_k^{\text{P}} \\ 0, & i \notin \mathcal{S}_k^{\text{P}} \end{cases} \end{aligned} \quad (42)$$

$$\begin{aligned} \mathbb{E} \left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\} & = \sum_{i \in \mathcal{S}_k^{\text{P}}} p_i \left| \text{tr}(\mathbf{\Theta}_{lk, li, t_{\tau_p} t}) \right|^2 \\ & + \text{tr} \left\{ \left[\sigma_z^2 \mathbf{I}_N + \sum_{i=1}^K p_i \beta_{li}^{\text{MP}} \mathbf{R}_{li} + \sum_{i=1}^K p_i |\alpha_{li}^{\text{LP}}|^2 \mathbf{h}_{li, t}^{\text{LP}} (\mathbf{h}_{li, t}^{\text{LP}})^H \right] \right. \\ & \cdot \left[\mathbf{d}_{lk} \mathbf{d}_{lk}^H + p_k \mathbf{\Delta}_{lk, t_k^{\text{P}} t_{\tau_p}} \mathbf{\Psi}_{l, t_k^{\text{P}} t_{\tau_p}} \mathbf{\Delta}_{lk, t_k^{\text{P}} t_{\tau_p}}^H \right] \Big\} \\ & + 2\text{Re} \left\{ \mathbf{d}_{lk}^H \sum_{i \in \mathcal{S}_k^{\text{P}}} p_i \mathbf{\Theta}_{lk, li, t_{\tau_p} t} \alpha_{li}^{\text{LP}} \mathbf{h}_{li, t}^{\text{LP}} \right\} \end{aligned} \quad (43)$$

and (44), as shown at the bottom of the page. Here, the mismatch term is denoted by $\mathbf{d}_{lk} \triangleq \mathbf{b}_{lk, t_k^{\text{P}} t_{\tau_p}} + (\hat{\alpha}_{lk}^{\text{LP}} - \alpha_{lk}^{\text{LP}}) \mathbf{h}_{lk, t_{\tau_p}}^{\text{LP}}$ and the cross-covariance matrix $\mathbf{\Theta}_{lk, li, t_{\tau_p} t}$ is defined as

$$\begin{aligned} \mathbf{\Theta}_{lk, li, t_{\tau_p} t} & \triangleq \text{Cov} \left\{ \hat{\mathbf{h}}_{lk, t_{\tau_p}}^{\text{Im}} - \hat{\mathbf{h}}_{lk, t_{\tau_p}}^{\text{Pe}}, \mathbf{h}_{li, t} | \mathcal{G}_l \right\} \\ & = \begin{cases} -\sqrt{p_k p_i} \beta_{li}^{\text{MP}} \rho_{li, t_k^{\text{P}} t_{\tau_p}} \rho_{li, t_{\tau_p} t} \mathbf{\Delta}_{lk, t_k^{\text{P}} t_{\tau_p}} \mathbf{R}_{li}, & i \in \mathcal{S}_k^{\text{P}} \\ \mathbf{0}_{N \times N}, & i \notin \mathcal{S}_k^{\text{P}}. \end{cases} \end{aligned} \quad (45)$$

The expectations and covariance in (42)–(45) are taken over RVs $\mathbf{h}_{li, t}^{\text{MP}}$, $s_k(t)$, and $\mathbf{z}_{l, t}$.

Proof: Please refer to Appendix C.

Theorem 3 establishes a rigorous closed-form lower bound for the conditional uplink SINR, which systematically characterizes the coupled effects of imperfect statistical CSI, channel aging, spatial correlation, and multiuser interference. Note that, rather than treating the statistical parameter errors as

generic additive noise, (41) explicitly quantifies the interference penalty ISC_{lk} incurred by the statistical mismatch in the receive combiner. This formulation directly relates the reliability of the environmental channel map to the achievable communication performance, thereby providing a mathematically tractable foundation for the subsequent transmission design. To gain deeper physical insight into the fundamental performance limits imposed by the channel map fidelity, we analyze the asymptotic behavior of the derived SINR as the number of BS antennas N grows without bound.

Corollary 1 (Massive MIMO Regime): As the number of BS antennas grows large ($N \rightarrow \infty$), the conditional uplink SINR of the k th UE converges to the following limit

$$\lim_{N \rightarrow \infty} \text{SINR}_{lk}(t) | \mathcal{G}_l = \frac{p_k \left| \alpha_{lk}^{\text{LP}} \right|^2 e^{j\phi_{lk}(t)} + \rho_{lk, t_{\tau_p} t} q_{lk, lk}^{(1)} \right|^2}{\sum_{i \in \mathcal{S}_k^{\text{P}} \setminus \{k\}} p_i \left| \rho_{li, t_{\tau_p} t} q_{lk, li}^{(1)} \right|^2 + I_{lk}^{\text{Asy}}} \quad (46)$$

where $\phi_{lk}(t)$ denotes the deterministic phase shift of the LoS path induced by terminal mobility between the time instant t_{τ_p} and t . The asymptotic normalized cross-covariance is defined as $q_{lk, li}^{(1)} \triangleq \lim_{N \rightarrow \infty} \frac{1}{N} \text{tr}(\mathbf{Q}_{lk, li, t_{\tau_p} t})$, the asymptotic interference penalty is denoted by $I_{lk}^{\text{Asy}} \triangleq \lim_{N \rightarrow \infty} \frac{1}{N^2} \mathbb{E} \left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\}$. It is nonnegative and becomes strictly positive when residual channel-map errors persist in the massive MIMO regime.

Proof: Please refer to Appendix D.

Corollary 1 reveals the fundamental performance bottlenecks of map-assisted massive MIMO systems as $N \rightarrow \infty$. While channel hardening asymptotically eliminates combining uncertainty and additive noise, the channel aging effect inherently persists, attenuating signal power via $\rho_{lk, t_{\tau_p} t}$ and inducing a phase misalignment $\phi_{lk}(t)$. Furthermore, unlike conventional systems limited solely by pilot contamination, (46) proves that map-induced statistical errors generate an irreducible interference floor I_{lk}^{Asy} . Consequently, deploying massive antenna arrays cannot compensate for a low-fidelity channel map.

B. Trade-Off Analysis Under Imperfect Statistical CSI

In this subsection, we investigate the block length design and the fundamental trade-off relationships among decoding error probability, transmission latency, and ADR in HRLLC systems. The practical operational logic of the BS under imperfect statistical CSI is systematically modeled based on the certainty-equivalence principle. According to short-blocklength information theory, the actual uplink ADR $\bar{R}_{lk}$

$$\begin{aligned} \Xi_{lk, li, t_{\tau_p} t}^{\text{Sq}} & \triangleq \mathbb{E} \left\{ \left| \left(\hat{\mathbf{h}}_{lk, t_{\tau_p}}^{\text{Pe}} \right)^H \mathbf{h}_{li, t} \right|^2 | \mathcal{G}_l \right\} = \text{tr} \left\{ \left[|\alpha_{lk}^{\text{LP}}|^2 \mathbf{h}_{lk, t_{\tau_p}}^{\text{LP}} (\mathbf{h}_{lk, t_{\tau_p}}^{\text{LP}})^H + \mathbf{Q}_{lk, lk, t_{\tau_p} t} \right] \left[|\alpha_{li}^{\text{LP}}|^2 \mathbf{h}_{li, t}^{\text{LP}} (\mathbf{h}_{li, t}^{\text{LP}})^H + \beta_{li}^{\text{MP}} \mathbf{R}_{li} \right] \right\} \\ & + \begin{cases} \left| \rho_{li, t_{\tau_p} t} \right|^2 \left| \text{tr}(\mathbf{Q}_{lk, li, t_{\tau_p} t}) \right|^2 + 2\text{Re} \left\{ \alpha_{lk}^{\text{LP}} \alpha_{li}^{\text{LP}} \rho_{li, t_{\tau_p} t} \left(\mathbf{h}_{lk, t_{\tau_p}}^{\text{LP}} \right)^H \mathbf{h}_{li, t}^{\text{LP}} \text{tr}(\mathbf{Q}_{lk, li, t_{\tau_p} t}) \right\}, & i \in \mathcal{S}_k^{\text{P}} \\ 0, & i \notin \mathcal{S}_k^{\text{P}}. \end{cases} \end{aligned} \quad (44)$$

of the k th UE in the l th cell employing FBL transmission can be approximated as [6]

$$\bar{R}_{lk} = C_{lk} - \frac{Q^{-1}(\epsilon)}{\tau_c} \sqrt{V_{lk}} \quad (47)$$

where ϵ denotes the target decoding error probability and $Q^{-1}(\cdot)$ denotes the inverse Q -function. The effective channel capacity C_{lk} and the aggregated channel dispersion V_{lk} over the data transmission duration are respectively given by [6]

$$C_{lk} = \frac{1}{\tau_c} \sum_{\eta=\tau_p+1}^{\tau_c} \log_2 \left(1 + \text{SINR}_{lk}(t) \right) \quad (48)$$

$$V_{lk} = (\log_2 e)^2 \sum_{\eta=\tau_p+1}^{\tau_c} \left[1 - \frac{1}{(1 + \text{SINR}_{lk}(t))^2} \right] \quad (49)$$

where $\text{SINR}_{lk}(t)$ is the conditional SINR derived in (41).

As communication time t progresses, the channel estimate acquired at t_{τ_p} becomes increasingly outdated due to channel aging, which generally reduces the effective coherent gain and degrades the instantaneous ADR. This reveals a fundamental trade-off in determining the transmission block length. An excessively short block length causes pilot symbols to consume a disproportionate share of time resources, reducing the average ADR. Conversely, an overly long block length exacerbates channel estimation inaccuracy over the data transmission phase, which likewise degrades the average ADR. A careful optimization of the block length is therefore imperative. While a heuristic pilot-spacing optimization scheme was proposed in [23] to alleviate the impact of channel aging, it does not account for the fundamental performance limits and the penalty terms inherent to FBL transmission in HRLLC systems. The block length optimization problem is formulated as

$$\text{P1: } \underset{\tau_c \in \mathbb{Z}^+}{\text{maximize}} \quad \bar{R}_{lk} \quad (50a)$$

$$\text{subject to } \tau_p + 1 \leq \tau_c \leq \tau_{th} \quad (50b)$$

where $\tau_{th} = \lfloor t_{th}/T_{\text{sym}} \rfloor$ enforces the latency constraint t_{th} .

Problem P1 is a bounded one-dimensional nonlinear integer program. Given the stringent latency requirements of HRLLC systems, the feasible range of the block length τ_c is tightly restricted. In FBL information theory, the average ADR exhibits quasi-concave (unimodal) behavior with respect to the block length, initially increasing to a peak before decreasing. Exploiting this unimodality and the limited variable range, the optimal block length can be determined by a one-dimensional search over a bounded integer interval. Since the feasible range is small in latency-constrained HRLLC systems, the associated computational cost is limited. The resulting computational complexity is $\mathcal{O}(\log_2(\tau_{th} - \tau_p))$, rendering this approach well-suited for latency-critical deployments.

When the acquired statistical CSI is perfect, τ_c^* attains the global optimum. Under imperfect channel maps, however, the BS schedules transmission using a τ_c^* optimized for the mismatched perceived SINR. When this scheduled block length is applied to the actual physical channel, the resulting true ADR $\bar{R}_{lk}$ inevitably deviates from the optimum, incurring a quantifiable performance penalty. This demonstrates the

degradation induced by the BS's overconfidence in a low-fidelity channel map.

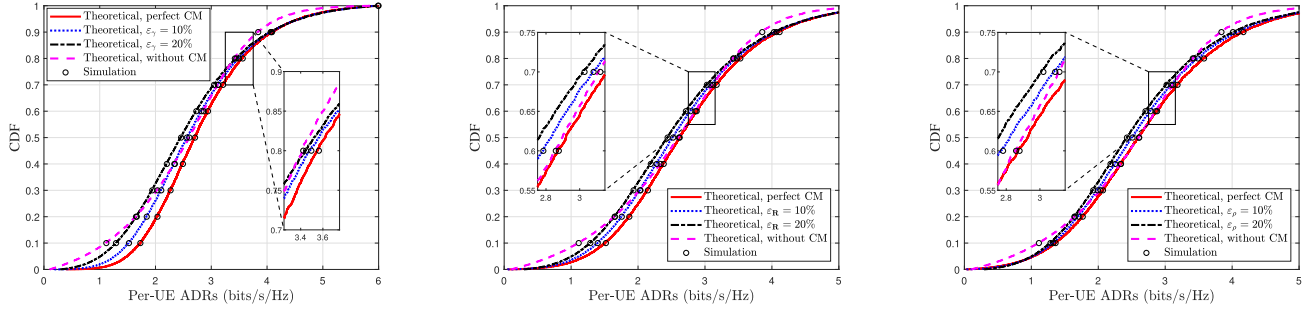
V. RESULTS AND ANALYSIS

In this section, we evaluate the performance of channel map-assisted HRLLC systems and provide a comparative analysis between systems with and without channel maps. Unless otherwise specified, the simulation parameters are as follows. The number of UEs in each cell is set as 10. The antenna number per BS is $N = 64$. The carrier frequency, system bandwidth, and subcarrier bandwidth are set to $f_c = 5.9$ GHz, $B = 20$ MHz, and $f_{\text{sub}} = 60$ kHz, respectively. The block length is set to $\tau_c = \tau_c^*$ via a one-dimensional search algorithm, and the pilot sequence length is $\tau_p = 10$. The UE speed is configured to $v_{li} = 10$ m/s. The channel map error level is set as 10%. The noise power spectral density is -174 dBm/Hz. The transmit powers of the i th UE are set to $p_i = 0.1$ W. Channel parameters are set as $M_{\text{path}} = 20$, $\kappa_\theta = 33$, and $\lambda_p = 0.2$. Throughout the simulations and analysis, “CM” and “without CM” denote the systems employing channel maps and operating without channel maps, respectively.

To examine the impact of individual CSI types on HRLLC performance, the cumulative distribution functions (CDFs) of per-UE ADRs obtained by exclusively utilizing channel maps to acquire the LoS indicator γ_{li} , the channel SCCF $\mathbf{R}_{li}$, and the channel TACF $\rho_{li,t_0,t}$ are presented in Fig. 3 (a), Fig. 3 (b), and Fig. 3 (c), respectively. Fig. 3 (a) demonstrates that HRLLC systems utilizing LoS indicator information with error levels of $\epsilon_\gamma = 0\%$, 10% , and 20% achieve average per-UE ADR improvements of 10.58%, 5.65%, and 1.05% relative to those without channel maps, respectively. Fig. 3 (b) reveals that the average per-UE ADR improves by 6.68%, 3.28%, and 0.53% when utilizing perfect, 10% error-level, and 20% error-level channel maps to obtain the channel SCCF, respectively. Fig. 3 (c) illustrates that HRLLC systems using perfect and 10% error-level, and 20% error-level channel maps to acquire TACF information achieve performance gains of 5.42%, 2.44%, and 0.13% respectively. These results show that exploiting the LoS indicator information yields the largest performance improvement, followed by the SCCF and then the TACF information.

The CDFs of per-UE ADRs under different channel map error levels are presented in Fig. 4. As expected, increasing map-error levels leads to the monotonic degradation in per-UE ADRs. Specifically, the average per-UE ADRs of HRLLC systems utilizing perfect, 10% error-level, and 20% error-level channel maps are 3.04, 2.85, and 2.59 bits/s/Hz, respectively. For reference, systems without channel maps achieve an average per-UE ADR of 2.54 bits/s/Hz, which is only marginally below that of the 20% error-level channel map. Relative to this baseline, the enhancements achieved by the perfect, 10% error-level, and 20% error-level channel maps are 19.58%, 12.03%, and 1.98%, respectively.

Fig. 5 presents the CDFs of per-UE ADRs under different UE speeds and block length configurations. For reference, the fixed block length $\tau_c = \tau_{th}$, commonly adopted in block-fading channel models, is included as a benchmark. The average per-UE ADRs of systems employing $\tau_c = \tau_{th}$



(a) CDFs of per-UE ADRs with different channel map-provided $\hat{\gamma}_{li}^{CM}$. (b) CDFs of per-UE ADRs with different channel map-provided $\hat{\mathbf{R}}_{li}^{CM}$. (c) CDFs of per-UE ADRs with different channel map-provided $\hat{\rho}_{li,t_0}^{CM}$.

Fig. 3. CDFs of per-UE ADRs with different channel map-provided (a) $\hat{\gamma}_{li}^{CM}$, (b) $\hat{\mathbf{R}}_{li}^{CM}$, (c) $\hat{\rho}_{li,t_0}^{CM}$.

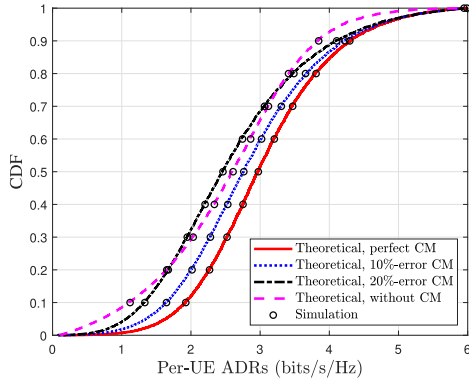


Fig. 4. CDFs of per-UE ADRs with different channel map-provided $\hat{\gamma}_{li}^{CM}$, $\hat{\rho}_{li,t_0}^{CM}$, and $\hat{\mathbf{R}}_{li}^{CM}$.

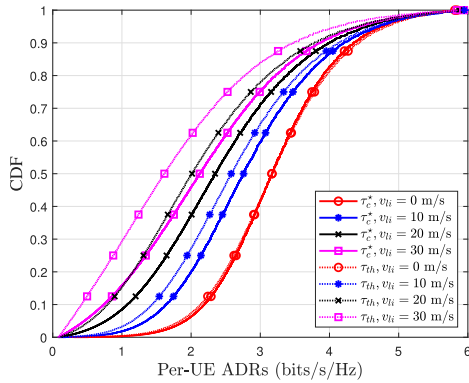


Fig. 5. CDFs of per-UE ADRs under different UE speeds.

at UE speeds $v_{li} \in \{0, 10, 20, 30\}$ m/s are 3.22, 2.69, 2.17, and 1.81 bits/s/Hz, respectively. By comparison, the optimized block length τ_c^* yields ADR improvements of 0.06%, 5.82%, 13.26%, and 23.41% at the corresponding UE speeds, respectively. These results demonstrate that block length optimization provides substantial performance gains in high-mobility scenarios, while its benefit is negligible in low-mobility conditions.

Fig. 6 illustrates the average per-UE ADR over the transmission block under different UE speeds. The average per-UE ADR initially increases and subsequently decreases in mobility scenarios, whereas it increases monotonically in the static scenario. In mobility scenarios, the block length τ_c^* obtained by solving Problem P1 via the one-dimensional search closely

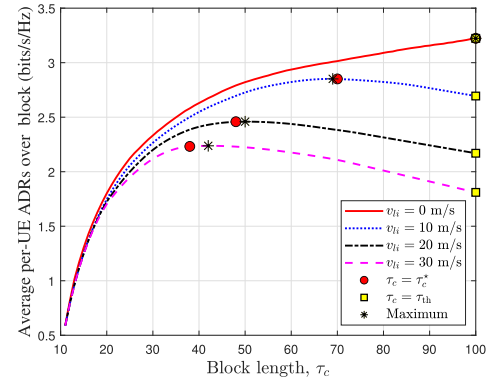


Fig. 6. Average per-UE ADRs over the transmission block under different UE speeds.

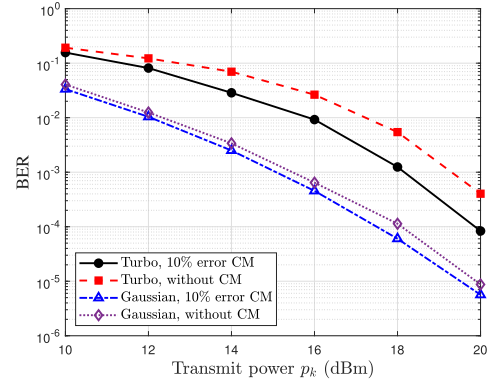


Fig. 7. BER versus transmit power under different coding schemes.

approximates the true optimum, and can thus be regarded as near-optimal. The discrepancy between τ_c^* and the true optimum is attributable to the map-error levels, and this discrepancy grows under higher UE mobility.

To evaluate system reliability under FBL transmission, Fig. 7 presents the bit error rate (BER) versus transmit power under different channel map conditions. Here, the performance of a practical Turbo coding scheme is benchmarked against the theoretical Gaussian codebook. The BER under the Gaussian codebook represents a theoretical lower bound that is unattainable in practice. The BER improvement conferred by the channel map is further demonstrated. Compared to the system without channel maps, the channel map-based system effectively suppresses the interference from imperfect prior CSI, even under a 10% error rate. These results validate

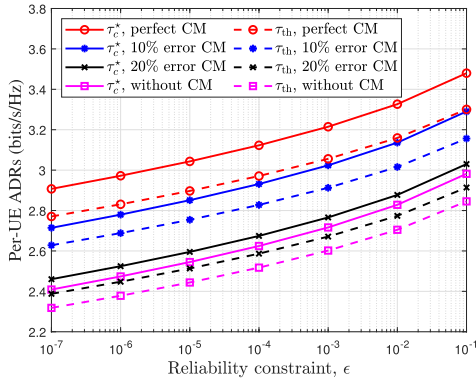


Fig. 8. Average per-UE ADRs over the transmission block under different error-level channel maps providing LoS indicator, channel SCCF, and channel TACF information.

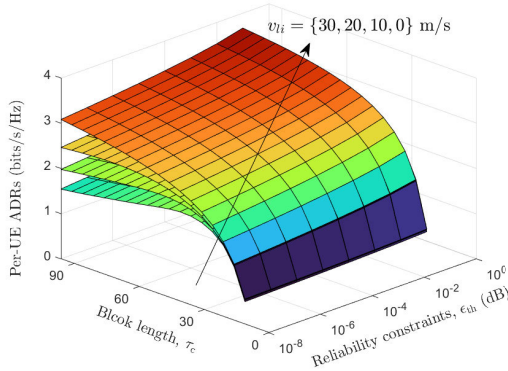


Fig. 9. Trade-off between per-UE ADRs and block length under different reliability constraints.

that exploiting environmental prior information fundamentally enhances the reliability of HRLLC transmissions.

Fig. 8 illustrates the per-UE ADR as a function of the reliability constraint under different block lengths and channel map error levels. More stringent reliability constraints reduce the achievable throughput, whereas relaxed constraints permit the system to sustain higher ADRs, revealing a fundamental reliability-throughput trade-off. Furthermore, the performance gain from constraint relaxation increases monotonically with the degree of relaxation. The ADR enhancements due to constraint relaxation remain nearly uniform across different channel map error levels, indicating that the reliability-throughput trade-off is largely insensitive to map accuracy.

Fig. 9 evaluates the trade-off among per-UE ADR, reliability, and the block length under given UE mobility conditions. As ϵ is relaxed or τ_c increases, the per-UE ADR improves because the finite-blocklength penalty becomes less severe. However, the ADR decreases as the UE speed increases from 0 m/s to 30 m/s, which reflects the negative impact of mobility-induced channel aging. The performance gap across different UE speeds becomes pronounced under stringent reliability constraints and short block lengths.

VI. CONCLUSION

In this paper, we have presented a channel map-based CSI acquisition method that utilizes UE positions, BS locations, and multipath information to improve the accuracy of statistical CSI. The tightness of the derived expressions has

been confirmed through Monte Carlo simulations. Results have demonstrated that, compared with systems relying on conventional CSI acquisition, HRLLC systems based on channel maps achieve ADR gains that vary with the channel map error rate. Specifically, the corresponding performance enhancements are 19.58 %, 12.02 %, and 1.98 % for channel maps with error rates of 0 %, 10 %, and 20 %, respectively. Among the considered uses of channel maps, exploiting LoS indicator information yielded the largest performance improvement, followed by the use of SCCF information, while TACF-based enhancement provided more modest gains. Furthermore, block-length optimization substantially improved the trade-off between reliability and ADR, particularly in high-mobility scenarios.

APPENDIX A PROOF OF THEOREM 1

According to the orthogonality principle of the MMSE estimator, the unconditional NMSE of a target parameter x and its conditional mean estimator $\hat{x}$ is given by $\text{NMSE}(x, \hat{x}) = 1 - \mathbb{E}\{\hat{x}^2\} / \mathbb{E}\{x^2\}$. Furthermore, applying the law of total expectation over the marginal distribution of the channel map observation yields $\mathbb{E}\{\Pi_{li}\} = P_{li}^L$. For the LoS component α_{li}^{LP} , invoking the moment generating function of the log-normal distribution, the true second moment is given by

$$\mathbb{E}\left\{\left|\alpha_{li}^{LP}\right|^2\right\} = P_{li}^L \frac{K_{li}^R}{K_{li}^R + 1} \exp\left(-\xi_{dB} \chi_{li}^L + \frac{\xi_{dB}^2 \sigma_L^2}{2}\right). \quad (51)$$

Similarly, the second moment of (14) is

$$\mathbb{E}\left\{\left|\hat{\alpha}_{li}^{LP,CM}\right|^2\right\} = \Phi_{li} \frac{K_{li}^R}{K_{li}^R + 1} \exp\left(-\xi_{dB} \chi_{li}^L + \frac{\xi_{dB}^2 \sigma_L^2}{4}\right) \quad (52)$$

Taking the ratio of (52) to (51) directly yields (19).

For the NLoS component β_{li}^{MP} , using the definitions of A_{li} and B_{li} , the true unconditional expected power $\mathbb{E}\{|\beta_{li}^{MP}|^2\}$ can be expressed as

$$\mathbb{E}\left\{|\beta_{li}^{MP}|^2\right\} = P_{li}^L A_{li}^2 e^{\xi_{dB}^2 \sigma_L^2} + (1 - P_{li}^L) B_{li}^2 e^{\xi_{dB}^2 \sigma_N^2} \quad (53)$$

According to $\mathbb{E}\{\Pi_{li}\} = P_{li}^L$ and $\mathbb{E}\{\Pi_{li}^2\} = \Phi_{li}$, we have

$$\begin{aligned} \mathbb{E}\left\{\left|\hat{\beta}_{li}^{MP,CM}\right|^2\right\} &= \mathbb{E}\left\{\left|\Pi_{li} A_{li} + (1 - \Pi_{li}) B_{li}\right|^2\right\} \\ &= \Phi_{li} (A_{li} - B_{li})^2 + 2P_{li}^L B_{li} (A_{li} - B_{li}) + B_{li}^2 \end{aligned} \quad (54)$$

which constitutes precisely the numerator of the subtracted term in (20). The expectations in (51)–(54) are taken over RVs F_{li}^L , F_{li}^N , and γ_{li} . To simplify the notation, we additionally define $q_{li,1} \triangleq P_{li}^L (1 - \epsilon_\gamma) + (1 - P_{li}^L) \epsilon_\gamma$ and $q_{li,0} \triangleq P_{li}^L \epsilon_\gamma + (1 - P_{li}^L) (1 - \epsilon_\gamma)$. Then, according to Bayes' rule, we have

$$\Pi_{li,1} = \mathbb{P}(\gamma_{li} = 1 \mid \hat{\gamma}_{li}^{CM} = 1) = \frac{P_{li}^L (1 - \epsilon_\gamma)}{q_{li,1}} \quad (55)$$

$$\Pi_{li,0} = \mathbb{P}(\gamma_{li} = 1 \mid \hat{\gamma}_{li}^{CM} = 0) = \frac{P_{li}^L \epsilon_\gamma}{q_{li,0}}. \quad (56)$$

Furthermore, based on $\Phi_{li} = q_{li,1} \Pi_{li,1}^2 + q_{li,0} \Pi_{li,0}^2$, we can obtain (21). By substituting (53) and (54) into the NMSE definition, the proof of Theorem 1 is completed.

APPENDIX B PROOF OF THEOREM 2

According to the definition of NMSE, the conditional NMSE is formulated as

$$\begin{aligned} \text{NMSE}(\mathbf{h}_{lk,t,\tau_p}, \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} | \mathcal{G}_l) \\ = \frac{1}{N[\alpha_{lk}^{\text{LP}}]^2 + \beta_{lk}^{\text{MP}}} \left\{ \left\| \mathbb{E}\left\{ \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} - \mathbf{h}_{lk,t,\tau_p} | \mathcal{G}_l \right\} \right\|^2 \right. \\ \left. + \text{tr}\left(\text{Cov}\left\{ \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} - \mathbf{h}_{lk,t,\tau_p} | \mathcal{G}_l \right\} \right) \right\}. \end{aligned} \quad (57)$$

By substituting the conditional mean of the estimated channel from (34) and the true channel mean $\mathbb{E}\{\mathbf{h}_{lk,t,\tau_p}\} = \alpha_{lk}^{\text{LP}} \mathbf{h}_{lk,t,\tau_p}^{\text{LP}}$ directly yields the squared norm term in (37), i.e., $\|(\hat{\alpha}_{lk}^{\text{LP}} - \alpha_{lk}^{\text{LP}}) \mathbf{h}_{lk,t,\tau_p}^{\text{LP}} + \mathbf{b}_{lk,t,\tau_p}^{\text{P}}\|^2$. Then, the zero-mean stochastic estimation error can be rewritten as

$$\begin{aligned} \mathbf{e}_{\text{sto}} &= \mathbf{h}_{lk,t,\tau_p}^{\text{MP}} - \sqrt{p_k} \hat{\rho}_{lk,t,\tau_p}^{\text{MP}} \hat{\mathbf{R}}_{lk} \hat{\Psi}_{l,t_k}^{-1} (\mathbf{y}_{l,t_k}^{\text{P}} - \bar{\mathbf{y}}_{l,t_k}^{\text{P}}) \\ &= \mathbf{e}_{\text{Pe}} + \sqrt{p_k} \Delta_{lk,t_k,t_{\tau_p}} (\mathbf{y}_{l,t_k}^{\text{P}} - \bar{\mathbf{y}}_{l,t_k}^{\text{P}}) \end{aligned} \quad (58)$$

where $\mathbf{e}_{\text{Pe}} \triangleq \mathbf{h}_{lk,t,\tau_p}^{\text{MP}} - \sqrt{p_k} \beta_{lk}^{\text{MP}} \rho_{lk,t_k,t_{\tau_p}} \mathbf{R}_{lk} \Psi_{l,t_k}^{-1} (\mathbf{y}_{l,t_k}^{\text{P}} - \bar{\mathbf{y}}_{l,t_k}^{\text{P}})$. According to the orthogonality principle, $\mathbf{e}_{\text{Pe}}$ is orthogonal to the observation $\mathbf{y}_{l,t_k}^{\text{P}} - \bar{\mathbf{y}}_{l,t_k}^{\text{P}}$. Therefore, the cross-covariance between $\mathbf{e}_{\text{Pe}}$ and the linear mismatch term $\sqrt{p_k} \Delta_{lk,t_k,t_{\tau_p}} (\mathbf{y}_{l,t_k}^{\text{P}} - \bar{\mathbf{y}}_{l,t_k}^{\text{P}})$ is zero. The conditional error covariance matrix is given by

$$\begin{aligned} \text{Cov}\left\{ \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} - \mathbf{h}_{lk,t,\tau_p} | \mathcal{G}_l \right\} \\ = \text{Cov}\{\mathbf{e}_{\text{Pe}}\} + p_k \Delta_{lk,t_k,t_{\tau_p}} \text{Cov}\{\mathbf{y}_{l,t_k}^{\text{P}}\} \Delta_{lk,t_k,t_{\tau_p}}^{\text{H}} \\ = \beta_{lk}^{\text{MP}} \mathbf{R}_{lk} - \mathbf{Q}_{lk,lk,t_{\tau_p}} + p_k \Delta_{lk,t_k,t_{\tau_p}} \Psi_{l,t_k}^{\text{H}} \Delta_{lk,t_k,t_{\tau_p}}^{\text{H}}. \end{aligned} \quad (59)$$

Applying the trace operator $\text{tr}(\cdot)$ to (59) and recalling that $\text{tr}(\mathbf{R}_{lk}) = N$ precisely provides the remaining terms in the numerator of (37). The expectations and covariances in (57)–(59) are taken over RVs $\mathbf{h}_{li,t}^{\text{MP}}$ and $\mathbf{z}_{l,t}$. This completes the proof of Theorem 2.

APPENDIX C PROOF OF THEOREM 3

To derive the closed-form lower bound for the conditional uplink SINR, we invoke the use-and-then-forget capacity bounding technique. Conditioned on $\mathcal{G}_l$, and using (40), the SINR is written as

$$\begin{aligned} \text{SINR}_{lk}(t) | \mathcal{G}_l &= \mathbb{E}\left\{ |\text{DS}_{lk}(t)|^2 | \mathcal{G}_l \right\} \left\{ P_{\text{Intf}} \right. \\ &\quad \left. + \mathbb{E}\left\{ |\text{AN}_{lk}(t)|^2 | \mathcal{G}_l \right\} + \mathbb{E}\left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\} \right\}^{-1} \end{aligned} \quad (60)$$

where the matched-MMSE interference power is $P_{\text{Intf}} \triangleq \mathbb{E}\left\{ |\text{CGU}_{lk}(t)|^2 | \mathcal{G}_l \right\} + \sum_{i \neq k} \mathbb{E}\left\{ |\text{MUI}_{lk}(t)|^2 | \mathcal{G}_l \right\}$. Then, we proceed to derive each component in (60). The power of the desired signal at the data transmission instant t is

$$\begin{aligned} \mathbb{E}\left\{ |\text{DS}_{lk}(t)|^2 | \mathcal{G}_l \right\} &= p_k \left| \mathbb{E}\left\{ (\hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Pe}})^{\text{H}} \mathbf{h}_{lk,t} \right\} \right|^2 \\ &= p_k |\Xi_{lk,lk,t_{\tau_p}t}|^2. \end{aligned} \quad (61)$$

According to the inner-product identity for complex random vectors, the inner expectation $\Xi_{lk,li,t_{\tau_p}t}$ is evaluated in closed form as (42). Using the definitions of $\text{CGU}_{lk}(t)$ and $\text{MUI}_{lk}(t)$, the matched-MMSE interference power is obtained as

$$\begin{aligned} P_{\text{Intf}} &= \sum_{i=1}^K p_i \mathbb{E}\left\{ |(\hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Pe}})^{\text{H}} \mathbf{h}_{li,t}|^2 \right\} - \mathbb{E}\left\{ |\text{DS}_{lk}(t)|^2 | \mathcal{G}_l \right\} \\ &= \sum_{i=1}^K p_i \Xi_{lk,li,t_{\tau_p}t}^{\text{Sq}} - p_k |\Xi_{lk,lk,t_{\tau_p}t}|^2. \end{aligned} \quad (62)$$

Based on the fourth-order moment theorem for jointly complex Gaussian vectors, the squared inner-product expectation $\Xi_{lk,li,t_{\tau_p}t}^{\text{Sq}}$ expands to (44). Since the noise $\mathbf{z}_{l,t}$ is independent of the channel estimates, the additive noise power is

$$\mathbb{E}\left\{ |\text{AN}_{lk}(t)|^2 | \mathcal{G}_l \right\} = \sigma_z^2 \left[|\alpha_{lk}^{\text{LP}}|^2 N + \text{tr}(\mathbf{Q}_{lk,lk,t_{\tau_p}}) \right]. \quad (63)$$

Substituting (39) into the definition of $\text{ISC}_{lk}(t)$ and exploiting the mutual independence of the data symbols and the additive noise, the interference-plus-spatial-correlation penalty power reduces to

$$\begin{aligned} \mathbb{E}\left\{ |\text{ISC}_{lk}(t)|^2 | \mathcal{G}_l \right\} &= \sum_{i=1}^K p_i \mathbb{E}\left\{ |(\hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} - \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Pe}})^{\text{H}} \mathbf{h}_{li,t}|^2 \right\} \\ &\quad + \sigma_z^2 \mathbb{E}\left\{ \left\| \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Im}} - \hat{\mathbf{h}}_{lk,t,\tau_p}^{\text{Pe}} \right\|^2 \right\}. \end{aligned} \quad (64)$$

The expectations in (60)–(64) are taken over RVs $\mathbf{h}_{li,t}^{\text{MP}}$, $s_k(t)$, and $\mathbf{z}_{l,t}$. Applying the fourth-order moment theorem to the inner-product expectation, (64) can be simplified to the final closed-form expression in (43).

APPENDIX D PROOF OF COROLLARY 1

To evaluate the asymptotic behavior as the number of antennas $N \rightarrow \infty$, we divide both the numerator and the denominator of the SINR expression in (41) by N^2 .

Firstly, based on the asymptotic orthogonality of the array steering vectors in massive MIMO, the normalized inner product of the LoS vectors satisfies $\lim_{N \rightarrow \infty} \frac{1}{N} (\mathbf{h}_{lk,t_{\tau_p}}^{\text{LP}})^{\text{H}} \mathbf{h}_{li,t}^{\text{LP}} = e^{j\phi_{lk}(t)}$ for $i = k$, and equals zero for $i \neq k$. Applying this limit to (42), the coherent signal gain converges to

$$\begin{aligned} \lim_{N \rightarrow \infty} \frac{1}{N} \Xi_{lk,li,t_{\tau_p}t} &= \begin{cases} |\alpha_{lk}^{\text{LP}}|^2 e^{j\phi_{lk}(t)} + \rho_{lk,t_{\tau_p}t} q_{lk,li}^{(1)}, & i = k \\ \rho_{li,t_{\tau_p}t} q_{lk,li}^{(1)}, & i \in \mathcal{S}_k^{\text{P}} \setminus \{k\} \\ 0, & \text{otherwise.} \end{cases} \end{aligned} \quad (65)$$

Furthermore, the channel hardening effect dictates that the normalized variance of the channel inner product strictly vanishes asymptotically, yielding $\lim_{N \rightarrow \infty} \frac{1}{N^2} \Xi_{lk,li,t_{\tau_p}t}^{\text{Sq}} = |\lim_{N \rightarrow \infty} \frac{1}{N} \Xi_{lk,li,t_{\tau_p}t}|^2$. Consequently, the $\text{CGU}_{lk}(t)$ for the target user k approaches zero

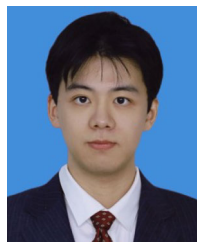
$$\lim_{N \rightarrow \infty} \frac{p_k}{N^2} \left(\Xi_{lk,lk,t_{\tau_p}t}^{\text{Sq}} - |\Xi_{lk,lk,t_{\tau_p}t}|^2 \right) = 0. \quad (66)$$

Finally, the additive noise power vanishes asymptotically under the $1/N^2$ normalization. Conversely, the statistical mismatch contained in $\text{ISC}_{lk}(t)$ exhibits a structural correlation

with the signal subspace, which causes its energy to scale as $\mathcal{O}(N^2)$. Thus, the normalized power converges to an error floor I_{lk}^{Asy} . Substituting these limits into the normalized form of (41) directly yields (46), which completes the proof.

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Yang Ou (Student Member, IEEE) received the B.E. degree in electronic information engineering from Nanjing University of Information Science and Technology, China, in 2022. He is currently pursuing the Ph.D. degree with the National Mobile Communications Research Laboratory, Southeast University, China. His research interests are performance analysis and design of HRLC systems, channel map-enabled transmission schemes, and impacts of channel characteristics on massive MIMO systems.



Jie Huang (Senior Member, IEEE) received the B.E. degree in information engineering from Xidian University, China, in 2013, and the Ph.D. degree in information and communication engineering from Shandong University, China, in 2018.

From October 2018 to October 2020, he was a Post-Doctoral Research Associate with the National Mobile Communications Research Laboratory, Southeast University, China, supported by the National Postdoctoral Program for Innovative Talents. From January 2019 to February 2020, he was a Post-Doctoral Research Associate with Durham University, U.K. Since March 2019, he has been a part-time Researcher with Purple Mountain Laboratories, China. Since November 2020, he has been an Associate Professor with the National Mobile Communications Research Laboratory, Southeast University. He has authored or co-authored more than 170 papers in refereed journals and conference proceedings. His research interests include millimeter wave, massive MIMO, reconfigurable intelligent surface channel measurements and modeling, electromagnetic information theory, and 6G wireless communications.

Dr. Huang received the IEEE Neal Shepherd Memorial Best Propagation Paper Award in 2024 and the Best Paper Awards from WPMC 2016, WCSP 2020, WCSP 2021, WCSP 2024, and IEEE ICCT 2025. He has served as an Editor for IEEE TRANSACTIONS ON GREEN COMMUNICATIONS AND NETWORKING and an Associate Editor for IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY. He has delivered more than 15 tutorials at international conferences, including IEEE Globecom and IEEE ICC.



Shuaifei Chen (Member, IEEE) received the B.S. degree in communication engineering and the Ph.D. degree in information and communication engineering from Beijing Jiaotong University, China, in 2018 and 2023, respectively.

From 2019 to 2020, he was a Visiting Scholar at the Department of Communication Systems, Linköping University, Sweden. From 2021 to 2022, he was a Visiting Scholar at the Division of Communication Systems, KTH Royal Institute of Technology, Sweden. He is currently a Research

Associate Professor with Purple Mountain Laboratories, Nanjing, China, and a Post-Doctoral Researcher with the National Mobile Communications Research Laboratory, School of Information Science and Engineering, Southeast University, Nanjing, where he has been an External Supervisor since January 2025. His research interests include signal processing and resource allocation for wireless communications, cell-free and ultra-massive MIMO, and digital twin-aided 6G multiple-antenna technologies. He was recognized as an Exemplary Reviewer for IEEE TRANSACTIONS ON COMMUNICATIONS in 2021 and IEEE COMMUNICATIONS LETTERS in 2023. He was selected for the Outstanding Postdoctoral Fellow Program in Jiangsu in 2024 and the Young Elite Scientists Sponsorship Program by Jiangsu Association for Science and Technology (JSAST) in 2025. He was a recipient of the Best Paper Award at IEEE ICC 2025. As a Principal Contributor, his project was selected for the 2025 Top Ten Scientific and Technological Advances in Information and Communications by China Institute of Communications (CIC). He served as the Special Session Co-Chair for IEEE SPAWC 2026 and a Lead Guest Editor for *Sensors*. He serves as an Associate Editor for the IEEE OPEN JOURNAL OF THE COMMUNICATIONS SOCIETY and an Editor for IEEE WIRELESS COMMUNICATIONS LETTERS. He also serves as a Technical Program Committee (TPC) Member for several conferences, including IEEE ICC and IEEE GLOBECOM.



Cheng-Xiang Wang (Fellow, IEEE) received the B.Sc. and M.Eng. degrees in communication and information systems from Shandong University, China, in 1997 and 2000, respectively, and the Ph.D. degree in wireless communications from Aalborg University, Denmark, in 2004.

He has been with Southeast University, Nanjing, China, as a Professor since 2018. He worked as the Dean of the School of Information Science and Engineering, Southeast University, from 2020 to 2026, and is currently a Vice President of Southeast University. He is also a Professor with Purple Mountain Laboratories, Nanjing. He was a Research Assistant with Hamburg University of Technology, Hamburg, Germany, from 2000 to 2001; a Visiting Researcher with Siemens AG Mobile Phones, Munich, Germany, in 2004; and a Research Fellow with the University of Agder, Grimstad, Norway, from 2001 to 2005. He was with Heriot-Watt University, Edinburgh, U.K., from 2005 to 2018, where he was promoted to a Professor in 2011. He has authored five books and more than 380 papers in refereed journals, including 35 highly cited papers. He has been granted over 50 invention patents, including four U.S. patents. He has also delivered 42 invited keynotes and 24 tutorials in international conferences. His current research interests include wireless channel measurements and modeling, 6G/B6G intelligent wireless networks, and electromagnetic information theory.

Dr. Wang is a member of the Academia Europaea (The Academy of Europe) and European Academy of Sciences and Arts (EASA); a fellow of the Royal Society of Edinburgh (FRSE), IET, and China Institute of Communications (CIC); an IEEE Communications Society Distinguished Lecturer in 2019 and 2020; and a Highly-Cited Researcher recognized by Clarivate Analytics (2017–2020) and in 2025. He was an Executive Editorial Committee Member of IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS from 2019 to 2025. He received the IEEE Neal Shepherd Memorial Best Propagation Paper Award in 2024 and 2026. He also received 21 Best Paper Awards from international conferences. He has served as an Editor for over 16 international journals, including IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS from 2007 to 2009, IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY from 2011 to 2017, and IEEE TRANSACTIONS ON COMMUNICATIONS from 2015 to 2017. He was a Guest Editor of the IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, IEEE TRANSACTIONS ON BIG DATA, and IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING. He has served as the TPC chair and the general chair for more than 30 international conferences.



El-Hadi M. Aggoune (Life Senior Member, IEEE) received the Ph.D. degree in electrical engineering from the University of Washington, Seattle, WA, USA.

He has worked at several universities in USA and abroad across various academic ranks, including the Endowed Chair Professor. He is listed as an Inventor on several patents assigned to universities and corporations, including The Boeing Company. He is also a registered Professional Engineer with the State of Washington. He has co-authored papers in IEEE and other prominent journals and conferences, while also serving on numerous editorial boards and technical committees. He is currently a Professor and the Director of the AI and Sensing Technologies Research Center, University of Tabuk, Tabuk, Saudi Arabia. His research interests include wireless communication, power systems, and artificial intelligence.

Dr. Aggoune received the IEEE Professor of the Year Award at the University of Washington. Previously, he directed a laboratory that received the Boeing Supplier Excellence Award.